

Short Paper

The Mechanism behind Understanding Anaphoric *One* and Noun Phrase Ellipsis: How Language Models Work

Nayoun Kim^{*,1}, JinYeong Bak², Yunyan Duan³, Ziyi Li¹, Sanskriti⁴, Kwangsu Kim^{**,4}

¹ Department of English Language and Literature/Sungkyunkwan University, Korea
nayoun@skku.edu; lydia.lzy@g.skku.edu

² College of Computing and Informatics - Department of Computer Science and Engineering/Sungkyunkwan University, Korea
jy.bak@skku.edu

³ Centre for Cognitive Science/Technical University of Darmstadt, Germany
yunyan.duan@tu-darmstadt.de

⁴ College of Computing and Informatics - Department of Computer Science and Engineering; Department of Artificial Intelligence/Sungkyunkwan University, Korea
sanskriti.srivastava07@gmail.com; kim.kwangsu@skku.edu

This study examines how language models access and recover linguistic information associated with the correct antecedents of two types of anaphoric elements: Anaphoric one and Noun Phrase Ellipsis (NPE), both of which require antecedent resolution for proper interpretation. In this study, we created manually annotated datasets (collectively referred to as AO-NPE datasets) from English Wikipedia, containing annotated instances of Anaphoric one and NPE. We assessed model performance by fine-tuning BERT and RoBERTa, and by evaluating GPT-4o mini, LLaMA 3.2, and Gemma-2 in a zero-shot setting. We conducted experiments using the forced-choice task and open-ended antecedent identification task on both humans and models with the same datasets. Notably, GPT-4o mini performed comparably to humans and outperformed other language models. However, results from both types of tasks revealed that language models performed better in identifying the antecedent for NPE than for Anaphoric one, a pattern opposite to that observed in human data. Linear distance and number agreement between antecedents and anaphoric elements also affected the performance, with language models more strongly influenced by these factors than humans. These findings suggest that the underlying mechanisms of anaphoric comprehension in language models may not fully align with those employed by humans: while humans flexibly integrate syntactic, semantic, and discourse infor-

* First author

** Corresponding authors

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mation, LLMs rely more heavily on surface-level cues such as linear distance, number agreement, and syntactic parallelism. Additionally, model performance dropped significantly in the open-ended identification task compared to the forced-choice task, while human performance remained relatively stable. This further suggests potential limitations in anaphoric processing.

1 Introduction

Anaphora resolution is fundamental to language comprehension. Anaphora occurs when an expression derives its meaning from a prior word or phrase, known as the antecedent (Huddleston and Pullum 2002; Llobart-Huesca 2002; Payne, Derbyshire, and Pullum 2013). The link between an anaphora and its antecedent is called a dependency, and resolving it can be challenging because the two elements may be separated by an unbounded distance (Linzen, Dupoux, and Goldberg 2016; Bacon and Regier 2019; Ryu and Lewis 2021; Lakretz et al. 2022). Nonetheless, anaphoric expressions are pervasive in everyday communication (Eckert and Strube 2000; Poesio, Stuckardt, and Versley 2016) and crucial for language processing tasks such as information retrieval and entity recognition (Sukthanker et al. 2020; Poesio et al. 2023).

While the resolution of common anaphoric expressions, such as pronouns (e.g., *he*, *she*, *it*), has been widely studied in both linguistics and computational linguistics, more linguistically complex structures, such as Anaphoric *one* and Noun Phrase Ellipsis (NPE), have been extensively examined in psycholinguistics and linguistics (Ross 1967; Jackendoff 1977; Lobeck 1995; Llobart-Huesca 2002; Lobeck 2006; Corver and Van Koppen 2011; Payne, Derbyshire, and Pullum 2013; Goldberg and Michaelis 2017; Kim, Brehm, and Yoshida 2019) but remain relatively underexplored in computational linguistics. Evaluating how language models resolve complex anaphoric structures and comparing their performance to that of humans shed light on different intelligent systems' capabilities and limitations in anaphora resolution.

In this study, we investigate whether large language models (LLMs) mirror human behavior in resolving two types of anaphora—Anaphoric *one* and NPE—and whether they capture the distinct linguistic properties involved in identifying their antecedents. We also examine how sensitive language models are to distance, morphological features, and parallelism strategies compared to humans.

1.1 Linguistic Analysis of Anaphoric *one* and NPE

Compared to common anaphoric expressions, understanding Anaphoric *one* can be particularly challenging because its antecedent may appear outside the same sentence and may not agree in number.

- (1) John has many tables. He decided to sell the red *one*.

In example (1), *one* is used anaphorically, deriving its meaning from the antecedent noun phrase (NP) “tables” in the previous sentence. The antecedent and Anaphoric *one* occur in different sentences and differ in number: “tables” carries the [+plural] number feature, whereas *one* has the [+singular] feature. Such morphological mismatches are permissible (Huddleston and Pullum 2002; Llobart-Huesca 2002; Culicover and Jackendoff 2005; Payne, Derbyshire, and Pullum 2013).

Understanding NPE is also challenging. In an NPE construction, the NP following certain elements, such as determiners (e.g., “some,” “any”; Lobeck 1995; Matthews 2014)

is omitted. The omitted NP is referred to as an ellipsis, and the determiner is referred to as an NPE licenser, as shown in (2).

- (2) Many students like syntax, though some [~~NP students~~] do not.¹

To understand NPE, one must first recognize that an NP has been elided (e.g., “students”) and then identify the antecedent that provides meaning to the elided NP. NPE is known to have an internal syntactic structure that parallels its antecedent; therefore, recovery of the grammatical, semantic, and morphological information associated with the antecedent is essential (Lobeck 1991; Merchant 2001; Kim, Brehm, and Yoshida 2019).

Anaphoric *one* and NPE exhibit distinct underlying linguistic properties. Both require antecedents for interpretation, but Anaphoric *one* is classified as a form of deep anaphora, relying on semantic or referential antecedents in the discourse representation that do not require a parallel linguistic antecedent (Hankamer and Sag 1976; Cornish 1996; Culicover and Jackendoff 2005). In contrast, NPE is surface anaphora, typically resolved through syntactic parallelism and access to both syntactic and semantic-discourse representations of the antecedent (Hankamer and Sag 1976; Sag and Hankamer 1984; Lobeck 1995; Merchant 2001; Kroll 2020).

Additionally, the interpretation of Anaphoric *one* and NPE may be influenced by morphological cues and distance. For *one*, number marking provides a clear cue to NP singularity, making it sensitive to agreement with its antecedent. For NPE, the ellipsis site is not overtly visible, which complicates antecedent access as distance increases (Martin and McElree 2011; Futrell, Mahowald, and Gibson 2015).

1.2 How Humans Process Anaphoric *one* and NPE: Psycholinguistic Evidence

The differing properties of NPE and Anaphoric *one* might suggest some differences in how their antecedents are accessed. Linguistically, Anaphoric *one* can be analyzed as a pro-form substituting for the head noun and can co-occur with modifiers (e.g., “the white hat and the red *one*”), referring to a previously mentioned discourse entity (Payne, Derbyshire, and Pullum 2013; Goldberg and Michaelis 2017). Given the linguistic properties of *one*, its referential antecedent is often identified through broader discourse-level information rather than through strict reliance on local syntactic constraints. The integration of semantic and discourse knowledge means the antecedent can be implicit or only partially specified rather than overtly stated; in such cases, meaning is often prioritized over structural constraints, which may occasionally result in incorrect antecedent predictions when the broader context overrides structural cues (Culicover and Jackendoff 2005; Kehler et al. 2008; Geiger and Xiang 2021).

In contrast, NPE antecedents are determined with sensitivity to grammatical, semantic, morphological, and discourse features. NPE poses a significant challenge for humans because it requires reconstructing the full syntactic structure of an antecedent while simultaneously accessing morphosyntactic, semantic, and discourse-level information. This process reflects active syntactic reconstruction at the ellipsis site, combined with the use of meaning and referential cues to interpret the ellipsis successfully (Frazier and Clifton 2005; Harris and Carlson 2017; Kroll 2020; Geiger and Xiang 2021; Rambelli et al. 2024).

¹ The strikethrough indicates that something has been omitted; in the case of NPE, the omitted element is the NP.

Recent psycholinguistic research has shown that humans access antecedents for Anaphoric *one* and NPE differently (Kim, Brehm, and Yoshida 2019; Kim and Frazier 2023). For Anaphoric *one*, readers primarily rely on discourse and morphological information, whereas for NPE, they also retrieve syntactic content associated with the antecedent, in addition to morphological and semantic information (Merchant 2001; Miller and Pullum 2014; Kroll 2020). Kim, Brehm, and Yoshida (2019) attributed the processing differences between Anaphoric *one* and NPE to readers' sensitivity to syntactic information specifying the antecedent, such as the head noun and its modifier.

1.3 How Language Models Resolve Anaphoric Dependencies

Computational approaches to anaphoric dependency have progressed from rule-based methods (Carbonell and Brown 1988) to data-driven statistical models (Aone and Bennett 1995; Ge, Hale, and Charniak 1998; Bos and Spenader 2011) and, more recently, neural network-based models (Clark and Manning 2016; Lee et al. 2017; Peters et al. 2018; Devlin et al. 2019). Early models relied on handcrafted rules and features (Hobbs 1978; Lappin and Leass 1994; Mitkov 2014; Lee et al. 2017), but their complexity and computational demands made them unsuitable for large datasets (Poesio et al. 2023). Later models, including neural models, addressed these limitations by learning from large-scale data and generalizing across languages and anaphoric elements (Lee et al. 2017; Devlin et al. 2019; Joshi et al. 2019; Wu et al. 2020).

Recent work has extended anaphora resolution to complex anaphoric elements such as ellipsis (Gardiner 2003; Bos and Spenader 2011; Liu, González Pellicer, and Gillick 2016; Khullar, Majmundar, and Shrivastava 2020; Khullar 2020; Hardt 2023; Testa, Chersoni, and Lenci 2023; Čavar et al. 2024). While neural models, particularly those leveraging BERT embeddings, significantly outperform earlier rule-based and statistical models (Khullar, Majmundar, and Shrivastava 2020), ellipsis remains a substantial challenge for contemporary NLP systems, including LLMs. For instance, Testa, Chersoni, and Lenci (2023) evaluated how two Transformer-based language models, GPT-2 and BERT, interpret elliptical sentences and recover missing elements in the ELLie dataset, which includes constructions such as Verb Phrase Ellipsis, Gapping, Pseudo-gapping, Sluicing, and Sluice-stranding. While both models performed well in recognizing typical events, they showed limited ability to reconstruct missing verbs. Similar difficulties were reported by Hardt (2023), who found that GPT-3 struggled with ellipsis compared to non-elliptical sentences. Čavar et al. (2024) evaluated NLP models on three tasks: (i) identifying the presence of ellipsis; (ii) detecting the place of an ellipsis; and (iii) reconstructing the elided content. They found that although GPT-4 showed some ability to generate missing material, overall performance remained poor across languages. Pre-trained BERT outperformed GPT-4 zero-shot models in detecting ellipsis locations.

It remains an open question which aspects of linguistic knowledge language models actually capture. Warstadt et al. (2020) introduced the Benchmark of Linguistic Minimal Pairs (BLiMP) to investigate the grammatical distinctions captured by Transformer-based language models, such as GPT-2 (Radford et al. 2019) and Transformer-XL (Dai et al. 2019), across syntactic, morphological, and semantic phenomena. These language models performed relatively well on morphologically related phenomena and ellipsis, but their performance remained significantly below human accuracy and revealed persistent challenges in other syntactic and semantic areas (see Rambelli et al. 2024, for an overview).

Although language models appear to learn syntactic rules (Liu et al. 2019) and LLMs appear to acquire linguistic knowledge during pre-training (Goldberg 2019; Hu, Chen, and Levy 2020; Linzen 2020; McCoy, Frank, and Linzen 2020; Warstadt et al. 2020), it remains unclear whether this reflects reliance on surface-level statistical cues from linear word sequences or the development of deeper hierarchical generalizations (Kim and Linzen 2020; McCoy, Frank, and Linzen 2020; Petty and Frank 2021; Kobzeva et al. 2025). Linzen, Dupoux, and Goldberg (2016) investigated whether language models could predict subject-verb agreement in sentences with dependency structures. For example, to understand sentence (3), the model must determine the correct verb form (singular or plural) based on the subject “the keys.”

(3) The keys to the cabinet ____.

In example (3), two NPs precede the verb: the NP “the keys,” which is [+plural], and the NP “the cabinet,” which is [+singular]. To complete a sentence, the model must first determine which of the NPs serves as the subject, thereby determining the number of the verb. This process requires distinguishing between the head (“the keys”) and the modifier (“to the cabinet”). However, models are known to incorrectly predict the number of the verb in cases where another NP (“the cabinet”) introduces singular number information. This suggests that language models may struggle to distinguish between different NPs within the sentence, revealing a broader difficulty in capturing hierarchical structure (Linzen, Dupoux, and Goldberg 2016; Arehalli and Linzen 2020; Linzen and Baroni 2021; Wilson, Zhou, and Frank 2023; Rambelli et al. 2024).

1.4 The Current Study

Given the importance of syntactic, semantic, and discourse accessibility, as well as superficial statistical cues such as proximity and syntactic parallelism strategies, in resolving Anaphoric *one* and NPE in both human language processing and computational models, this study investigates the capabilities and limitations of LLMs in processing such anaphoric dependencies and compares their performance to that of humans. We address the following research questions:

RQ 1. Do language models identify and access antecedents based on the distinct linguistic properties and recovery mechanisms required for Anaphoric *one* and NPE, and do they exhibit human-like patterns in resolving them?

RQ 2. To what extent are language models, compared to humans, influenced by distance, morphological features, and parallelism when identifying the antecedents of Anaphoric *one* and NPE?

We created a set of annotated datasets (collectively referred to as the **AO-NPE datasets**) specifically designed for the resolution of Anaphoric *one* and NPE. These datasets were developed to examine the sensitivity of language models to key factors such as linear distance and number agreement between the anaphoric elements and their antecedents, which are critical for accurate anaphora resolution. We evaluated model performance in a series of experiments, including a forced-choice task and an open-ended antecedent identification task employing BERT (Devlin et al. 2019), RoBERTa (Liu et al. 2019), as well as several LLMs, including *GPT-4o mini* (OpenAI 2024), *LLaMA 3.2* (Meta AI 2024), and *Gemma-2* (Gemma Team 2024). These models offer state-of-the-art performance among current language models and represent two key categories: encoder models specialized for language understanding (e.g., BERT and

RoBERTa) and foundational models designed for broader purposes beyond language understanding (e.g., LLMs).

This study makes several contributions. First, as noted earlier, only a limited number of datasets focused on ellipsis (Bos and Spenader 2011; Goldberg and Stubbs 2020; Anand, Hardt, and McCloskey 2021), and to our knowledge, few specifically address Anaphoric *one*. Moreover, existing evaluations of language models have predominantly concentrated on particular elliptical constructions. To address this gap, this study introduces a fine-grained and carefully curated annotated dataset for Anaphoric *one* and NPE and provides human baselines for comparison.

Second, this study contributes to the theoretical question of whether LLMs mirror human behavior in resolving Anaphoric *one* and NPE, and whether they capture the distinct linguistic properties involved in identifying their antecedents. While both NPE and *one* require an antecedent, they differ fundamentally in how it is recovered. Anaphoric *one*, a deep anaphora, relies primarily on semantic and referential information represented in discourse. In contrast, NPE is a surface anaphora that depends on syntactic, semantic, and discourse cues associated with the antecedent, with particular emphasis on syntactic information. These differences may point to variation in processing mechanisms, possibly modulated by factors such as morphological agreement, the distance between the anaphora and their antecedents, and parallelism (Gibson 1998; Van Dyke 2007; Martin and McElree 2008; Futrell, Mahowald, and Gibson 2015). Given that Anaphoric *one* and NPE are grammatically distinct despite their surface similarity, they offer a critical test of whether LLMs move beyond linear patterns to capture the syntactic, semantic, and discourse processes underlying anaphoric comprehension. In particular, when models misidentify an antecedent, what types of errors do they produce, and how do these error patterns align with those observed in humans?

To preview our results, we found that LLMs such as *GPT-4o mini* (OpenAI 2024) showed performance comparable to humans in identifying the correct antecedents of Anaphoric *one* and NPE. Both LLMs and humans were affected by linear distance and number agreement; however, humans exhibited less sensitivity to these factors. LLMs also relied more on parallelism strategies. An interesting finding was that humans found resolving NPE more challenging than resolving Anaphoric *one*, whereas certain models performed better on NPE. This could indicate potential differences in how humans and LLMs resolve Anaphoric *one* and NPE. These results were consistent across both task types. However, the models' performance in correctly identifying the antecedents of *one* and NPE dropped significantly in the open-ended identification tasks compared to the forced-choice tasks, while human performance remained stable.

2 Annotated AO-NPE Datasets

2.1 Datasets Created for the Anaphoric *one* and NPE Resolution Tasks

Given the lack of publicly available datasets for the Anaphoric *one* and NPE, we created our own AO-NPE datasets. First, we conducted a corpus search using Sketch Engine (Kilgariff et al. 2014) on the English Wikipedia corpus. For Anaphoric *one*, we searched for all instances of *one* using [lemma="one"], which returned approximately 4.5 million hits in the 61.6 billion-token corpus (English Wikipedia). We then manually examined 3,000 instances and found roughly 457 texts (15.23%) reflecting the anaphoric use of *one*. Extrapolating from this sample suggests that approximately 687,667 of the *one* tokens are anaphoric, which corresponds to 0.0011% of the total corpus. For NPE, we cannot directly identify occurrences in surface forms, since they involve empty elements.

Instead, we searched using common NPE licensors (e.g., “any,” “some”; Merchant 2001) and manually annotated 3,000 sentences, among which we identified 391 NPE instances. Extrapolating from this proportion to the roughly 6.1 million licensor tokens in the corpus suggests about 794,000 NPE instances (0.0013% of the total corpus).² This suggests that both Anaphoric *one* and NPE have relatively low frequencies in natural web texts, and are therefore infrequent in the training data of LLMs.

The texts were annotated using the standoff annotation method (Bos and Spenader 2011; Khullar, Majmundar, and Shrivastava 2020). A detailed description of the annotation procedure is provided in Section 2.2. In brief, for each item, we manually identified three key elements: the anaphoric elements (i.e., the Anaphoric *one* or the NPE licensor), the correct antecedent, and an additional NP that could plausibly be misidentified as the antecedent (i.e., the incorrect antecedent). Our datasets comprise five Excel sheets.

- **The G(eneral)A(naphoric)O(ne) corpus** includes 457 texts where either the correct or the incorrect antecedent can be close to or far from the Anaphoric *one* or agree or disagree with the Anaphoric *one* in the number feature.³
 - **The D(istance)F(actor)A(naphoric)O(ne) sub-corpus** includes 143 texts where the correct antecedent is always far from the Anaphoric *one* and is intervened by an incorrect one.
 - **The N(umber)F(actor)A(naphoric)O(ne) sub-corpus** includes 154 texts where the incorrect antecedent always matches the Anaphoric *one* in number, while the correct one does not.
- **The G(eneral)NPE corpus** includes 391 texts in which either the correct or incorrect antecedent is close to or far from the NPE licensor.
 - **The D(istance)F(actor)NPE sub-corpus** includes 164 texts in which the correct antecedent is always far from the NPE licensor and is intervened by an incorrect one.

Among the datasets, the GAO and GNPE corpora were created to evaluate the ability of different language models to correctly identify the antecedents of *one* and NPE. The sub-corpora were derived from these general corpora. The DFAO and DFNPE sub-corpora were designed to examine the influence of linear distance between *one*/NPE and their antecedents. According to previous research, the retrieval of fine-grained information such as lexical-semantic and structural properties of the antecedent is affected by distance (Gibson 1998; Wagers and Phillips 2014; Kim et al. 2020a). We therefore treated distance as a factor affecting the identification of the correct antecedent for *one* and NPE. The NFAO sub-corpus was designed to examine the influence of number agreement between *one*/NPE and their antecedents on successful anaphoric resolution.

² This estimate is necessarily approximate since the true frequency depends on the full set of possible licensors and their distributions; therefore, the actual number may be higher or lower.

³ The AO datasets include both NP-like uses of *one* that allow plural morphology (e.g., “those ones”) and DP-like uses of *one* (e.g., “you cannot create one”), which are DPs that do not pluralize (Lobeck 1995; Culicover and Jackendoff 2005; Huddleston and Pullum 2002). Thus, *one* functions either as a DP-level indefinite pronoun or as an NP-level pro-form (Harris 1946; Jackendoff 1977), and both types are represented in the corpus (86% vs. 14%). To complement our main analyses and examine anaphoric resolution under more controlled conditions, we constructed simplified AO-NPE datasets from the English Web 2021 (enTenTen21) corpus. The simplified dataset includes only NP-like uses of *one/s*, excludes number mismatches between the Anaphoric *one/s* and their antecedents, and retains only number-matching instances. In addition, we restricted the simplified dataset to cases in which the verb in the conjunct containing *one* differs from the preceding verb, and ensured that both candidates were equally plausible. Detailed dataset descriptions and the corresponding experimental results are provided in Appendix F and are available on GitHub (<https://github.com/nayoun1/Anaphoric-One-and-NPE>).

2.2 Annotation Method

Three trained annotators with linguistic backgrounds (each having completed at least two advanced courses) manually annotated the datasets using detailed guidelines. Disagreements were resolved through in-person discussion, and highly controversial cases were removed. All annotations were then reviewed by a linguist to ensure consistency and accuracy. Following prior work (Bos and Spenader 2011; Webster et al. 2018; Khullar, Majmundar, and Shrivastava 2020), we recorded the character offsets of the Anaphoric *one* and the NPE licenser, along with their candidate antecedents. Each candidate was marked as either the correct or incorrect antecedent of the anaphoric element using binary labels: “TRUE” (indicating a correct antecedent) or “FALSE” (indicating an incorrect antecedent). Table 1 presents an example of the annotation format from the GAO corpus.

Table 1

Annotation example from the GAO corpus. In column 2, Anaphoric *one* is shown in bold, and the correct antecedent is underlined.

column 1	2	3	4	5	6	7	8	9	19
text ID	text	<i>one/ones</i>	antecedent type	<i>one</i> -start character offset	<i>one</i> -end character offset	A	antecedent A-start character offset	antecedent A-end character offset	URL
1	(Text omitted for readability)... there is no <u>war</u> between us and you cannot create one !	one	linguistic	275	277	war	238	240	https://en.wikipedia.org/wiki/2009_shootings_of_Oakland_police_officers
column 10	11	12	13	14	15	16	17	18	
B	antecedent B-start character offset	antecedent B-end character offset	A-antecedent	B-antecedent	A-antecedent	B-antecedent	antecedent number	<i>one</i> -licensor category	
police	218	223	TRUE	FALSE	local	non-local	npsg	verb	

In our datasets, alongside the offsets of anaphoric elements and their candidate antecedents, as well as TRUE/FALSE labels indicating correct or incorrect antecedents, we included several additional columns (detailed in Table 2) to enrich the annotations.

Table 2

Specified columns added in the annotation (illustrated with the Anaphoric *one* example).

Raw text	(Text omitted)..., there is no war between us and you cannot create <i>one</i> !				
Columns	Anaphoric <i>one/ones</i> (C3)	Antecedent type (C4)	Distance/number matching between <i>one</i> and antecedents (C15/16)	Antecedent number (C17)	<i>One</i> -licensor (C18)
Example	One	linguistic	A: local; B: non-local A: match; B: mismatch	npsg	verb
Note	Marks the number feature of <i>one</i> , corresponding with Column 17, which labels the number feature of the correct antecedent.	Marks the type of the correct antecedent. “Linguistic” means the antecedent is within the same sentence as <i>one</i> , and “discourse” means it is found in previous context.	Labels the distance or number agreement between the antecedent options and <i>one</i> .	Labels the number feature of the true antecedent (“nppl” for plural NP, and “npsg” for singular NP), corresponding with the <i>one/ones</i> column (C3).	Marks the category of the phrase that introduces the Anaphoric <i>one</i> .

The DFAO sub-corpus is designed to examine the effect of linear distance between the Anaphoric *one* and its correct antecedent on model performance. It shares the same labeling scheme as the GAO corpus, with one key distinction: in the DFAO sub-corpus, the incorrect antecedent (either antecedent A or B) is always linearly closer to the Anaphoric *one* than the

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correct antecedent. For instance, in the instance shown in Table 3, antecedent B (“driver”) is the incorrect antecedent (marked as “FALSE” in Column 14) and appears closer to the Anaphoric *one* (marked as “local” in Column 16) than the correct antecedent, antecedent A (“excuse”), which is marked as “non-local” in Column 15.

Table 3
Annotation example from the DFAO sub-corpus.

column 1	2	3	4	5	6	7	8	9	19
text ID	text	<i>one/ones</i>	antecedent type	<i>one</i> -start character offset	<i>one</i> -end character offset	A	antecedent A-start character offset	antecedent A-end character offset	URL
1	(Text omitted for readability)... And even though I've heard every excuse every driver has ever made, I certainly hadn't heard of that one !	one	linguistic	236	238	excuse	168	173	https://en.wikipedia.org/wiki/Ayrton_Senna
column 10	11	12	13	14	15	16	17	18	
B	antecedent B-start character offset	antecedent B-end character offset	A-antecedent	B-antecedent	A-antecedent	B-antecedent	antecedent number	<i>one</i> -licensor category	
driver	181	186	TRUE	FALSE	non-local	local	npsg	det	

The NFAO sub-corpus follows a format nearly identical to that of the GAO and DFAO sub-corpus. However, since it was designed to examine the effect of number agreement between the Anaphoric *one* and its candidate antecedents on model performance, it differs in terms of whether the antecedent and Anaphoric *one* agree in number or not. Rather than indicating whether an antecedent is “local” or “non-local” to the Anaphoric *one*, the NFAO sub-corpus marks whether the antecedent “matches” or “mismatches” the Anaphoric *one* in number (as recorded in Columns 15 and 16). In all NFAO examples, the incorrect antecedent (either antecedent A or B) always agrees in number with the Anaphoric *one*, while the correct antecedent does not. For instance, in the example shown in Table 4, antecedent B (“world”) is the incorrect antecedent (marked as “FALSE” in Column 14) and agrees in number with the Anaphoric *one* (marked as “match” in Column 16). In contrast, the correct antecedent, antecedent A (“magic wands”), disagrees in number with the Anaphoric *one* (marked as “mismatch” in Column 15).

Table 4
Annotation example from the NFAO sub-corpus.

column 1	2	3	4	5	6	7	8	9	19
text ID	text	<i>one/ones</i>	antecedent type	<i>one</i> -start character offset	<i>one</i> -end character offset	A	antecedent A-start character offset	antecedent A-end character offset	URL
1	(Text omitted for readability)... “You'll need a lot of magic wands to change our world ; but let's make a start with this one !”	one	linguistic	373	375	magic wands	307	317	https://en.wikipedia.org/wiki/Silvio_Mantelli
column 10	11	12	13	14	15	16	17	18	
B	antecedent B-start character offset	antecedent B-end character offset	A-antecedent	B-antecedent	A-antecedent	B-antecedent	antecedent number	<i>one</i> -licensor category	
world	10	14	TRUE	FALSE	mismatch	match	nppl	det	

The format of the NPE corpora closely mirrors that of the AO corpora. However, since NPEs typically require linguistic antecedents (Hankamer and Sag 1976; Sag and Hankamer 1984), the column labeling antecedent type (Column 4 in the AO corpora) was excluded from the NPE annotation. The NPE annotation schema includes a total of 18 columns, as illustrated in Table 5.

Table 5
Annotation format of the NPE corpus (example from the GNPE corpus).

column 1	2	3	4	5	6	7	8	9
ID	text	licensor	licensor-start character offset	licensor-end character offset	A	antecedent A-start character offset	antecedent A-end character offset	B
2	Marty Links' Emmy Lou (November 14, 1966): "I suppose it's called 'mad money' because fathers always get so mad when you ask for some! "	some	129	132	fathers	86	92	money
column 10	11	12	13	14	15	16	17	18
antecedent B-start character offset	antecedent B-end character offset	A- antecedent	B- antecedent	A- antecedent	B- antecedent	antecedent number	licensor- category	URL
71	75	FALSE	TRUE	local	non-local	npsg	det	https://en.wikipedia.org/wiki/Marty_Links

The GNPE corpus and the DFNPE sub-corpus share the same labeling scheme, with one key distinction: in the DFNPE sub-corpus, the incorrect antecedent (either antecedent A or B) is always linearly closer to the NPE licensor than the correct antecedent. For example, in the instance shown in Table 6, antecedent A ("fathers") is the incorrect antecedent of the NPE (marked as "FALSE" in Column 12) and is closer to the licensor "some" (marked as "local" in Column 14) than antecedent B ("money"), which is the correct antecedent.

Table 6
Annotation example from the DFNPE sub-corpus.

column 1	2	3	4	5	6	7	8	9
ID	text	licensor	licensor-start character offset	licensor-end character offset	A	antecedent A-start character offset	antecedent A-end character offset	B
2	Marty Links' Emmy Lou (November 14, 1966): "I suppose it's called 'mad money' because fathers always get so mad when you ask for some! "	some	129	132	fathers	86	92	money
column 10	11	12	13	14	15	16	17	18
antecedent B-start character offset	antecedent B-end character offset	A- antecedent	B- antecedent	A- antecedent	B- antecedent	antecedent number	licensor- category	URL
71	75	FALSE	TRUE	local	non-local	npsg	det	https://en.wikipedia.org/wiki/Marty_Links

3 Experiments

In this study, we utilized our custom-annotated datasets for resolving Anaphoric *one* and NPE, which were introduced in previous sections. We investigated: (i) the extent to which language models accurately recover the information associated with the antecedents of Anaphoric *one* and NPE, given their distinct linguistic properties and mechanisms required for antecedent recovery, and (ii) the extent to which language models exhibit human-like patterns and rely on similar cues in identifying antecedents. We address these questions by comparing the performance of various language models on the GAO and GNPE corpora, as well as on the sub-corpora, against human performance in both open-ended and forced-choice identification tasks. In the forced-choice task setting, humans and language models chose the correct antecedent from two options. In contrast, in the open-ended antecedent identification task setting, no options were provided, and humans and models had to generate the antecedent directly.

3.1 Forced-choice Resolution Task

In the forced-choice experiment, humans and language models were expected to choose the correct antecedent from two provided options. We used our custom-annotated AO-NPE datasets to investigate human and model performance in Anaphoric *one* and NPE resolution. Alongside this, we included gendered and gender-neutral coreference resolution tasks to provide a reference point for language models' performance in typical anaphor resolution tasks. For this purpose, we incorporated the GAP (Webster et al. 2018) and WSC 273 (Levesque, Davis, and Morgenstern 2012) datasets. The GAP dataset included gendered pronouns such as “he” or “she,” while the WSC 273 dataset included Gender-neutral pronouns (mainly “it” or “they,” though also some examples of *one*). Both datasets consist of manually annotated English texts (see Tables 7 and 8).

Table 7

An example from the GAP dataset (Webster et al. 2018).

ID	Text	Pronoun	Pronoun-offset	A	A-offset	A-coref	B	B-offset	B-coref	URL
test-1	(Text omitted)... Bob Suter are Dehner's uncles. His cousin is Minnesota Wild's alternate captain Ryan Suter.	His	383	Bob Suter	352	FALSE	Dehner	366	TRUE	http://en.wikipedia.org/wiki/Jeremy_Dehner

Table 8

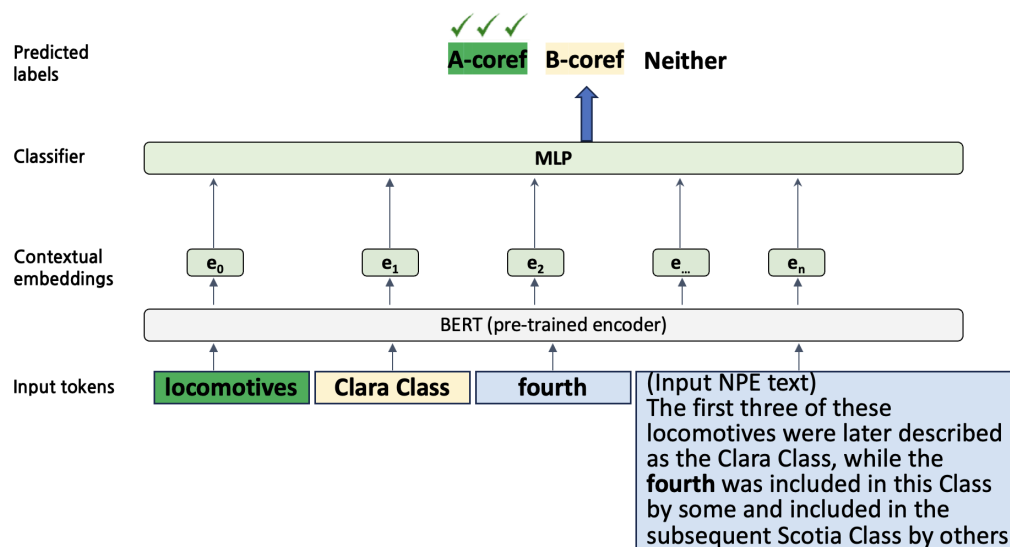
An example from the WSC 273 dataset (Levesque, Davis, and Morgenstern 2012)

text	pronoun	pronoun_loc	quote	quote_loc	options	source
The trophy doesn't fit into the brown suitcase because it is too large.	it	55	it is too large	55	["the trophy", "the suitcase"]	Hector Levesque

3.2 Models: BERT and RoBERTa

We incorporated a unified base training architecture across all the anaphor resolution tasks. Following Garcia et al. (2023), we used BERT (Devlin et al. 2019) and RoBERTa (Liu et al. 2019) models (large, uncased), fine-tuning them for our downstream text classification task of predicting the correct antecedent for each task. In this section, we describe our training process using BERT. For brevity and clarity, we adopted a similar training strategy for RoBERTa, adhering to the outlined principles. Figure 1 illustrates the detailed training architecture.

The process begins with the preparation of the input text, where each example consists of two possible antecedents (A and B), and one referring expression (P), such as “one” or “some.” In the case of Anaphoric *one* resolution, the referring expression is the Anaphoric *one*, such as “one.” For

**Figure 1**

The detailed training architecture across the forced-choice tasks (illustrated with an NPE text). In this example, the NPE licensor is “fourth,” the correct antecedent is “locomotives,” and the incorrect antecedent is “Clara Class.”

the NPE resolution, the referring expression is the NPE licensor (e.g., “some”). These elements are annotated in the input before being fed into the model. The BERT tokenizer converts the input text into tokens, and BERT encodes them into contextual representations. Then, a multi-layer perceptron (MLP) uses those representations to classify which candidate (A or B) the referring expression matches, or if neither matches. The output is a prediction: A-coref, B-coref, or Neither.

For example, as illustrated in Figure 1, the models were first trained with the input data such as the sentence (input “NPE” text) “The first three of these locomotives were later described as the Clara Class, while the **fourth** was included...,” where “locomotives” is candidate antecedent “A” (A-coref) and “Clara Class” is candidate antecedent “B” (B-coref), and the NPE licensor “fourth” is the referring expression. After processing the text, the model outputs “A-coref” (correct antecedent), “B-coref” (incorrect antecedent), or “Neither” (incorrect answer). Once trained, the model processes new sentences.

3.3 Models: LLMs

We also conducted zero-shot and fine-tuning experiments using various LLMs to identify the antecedents of anaphoric elements within a sentence. The models used in the experiments were OpenAI’s GPT-4o mini (OpenAI 2024), Meta’s LLaMA 3.2B-Instruct (Meta AI 2024), and Google’s Gemma 2.9B-Instruct (Gemma Team 2024). Figure 2 illustrates the zero-shot inference setup used to evaluate model performance on the anaphoric resolution tasks.

The input was preprocessed in a prompt format based on the datasets. The prompt consisted of a sentence containing an anaphoric element (either the Anaphoric *one* or an NPE licensor) and a pair of candidate nouns that could serve as the antecedent, in the following format:

Question: In the sentence ‘[]’, what does ‘[one]’ refer to?

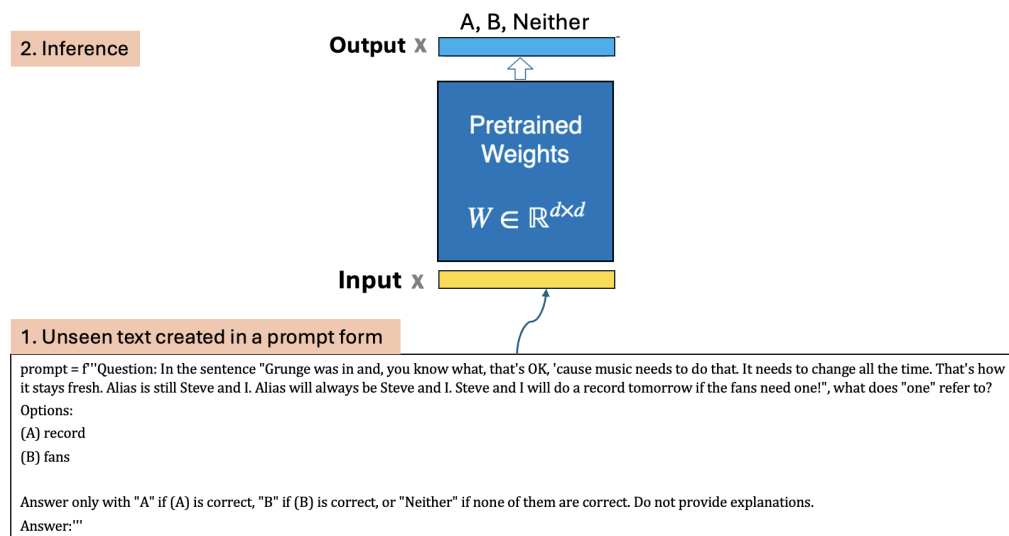
Options:

(A) [Noun A],

(B) [Noun B].

Answer only with ‘A’, ‘B’, or ‘Neither’.

Do not provide explanations.

**Figure 2**

Zero-shot forced-choice anaphora resolution via prompt-based inference.

We also conducted additional experiments using fine-tuned versions of the models. Details of the experimental setup and the results with these fine-tuned LLMs are provided in Appendix C.2 & 3.

3.4 Human Experiments with the Forced-choice task

We conducted an experiment using the same forced-choice setting to establish a baseline for human understanding of the antecedents of the Anaphoric *one* and NPE. Based on existing linguistics and psycholinguistics literature, we hypothesize that the accuracy of identifying the correct antecedent for NPE and Anaphoric *one* might differ due to their grammatical distinctions (Hankamer and Sag 1976; Merchant 2001; Llombart-Huesca 2002; Culicover and Jackendoff 2005; Payne, Derbyshire, and Pullum 2013; Goldberg and Michaelis 2017; Kim, Brehm, and Yoshida 2019), with NPE expected to be more challenging than the Anaphoric *one*. Furthermore, we predicted that factors such as number (for Anaphoric *one*) or distance (for both Anaphoric *one* and NPE) could influence the accuracy of correct antecedent identification.

3.4.1 Participants

135 self-reported native English-speaking participants (currently residing in the USA and Canada, and having learned English before the age of 5) were recruited through the crowdsourcing Prolific Academic Online Platform (Palan and Schitter 2018). Participants provided informed consent and were remunerated at approximately \$12/hour. No participant participated in the experiment more than once.

3.4.2 Materials and Procedure

The manually annotated GAO corpus contained 457 texts, while the GNPE corpus contained 391 texts. Each experiment consisted of 20-22 critical test items, resulting in a total of 23 sets for the Anaphoric *one* and 23 sets for NPE. For each experiment, 20-22 critical items were combined with two filler sentences to serve as attention check.

The experiment was deployed on the PC Ibex Farm web-based platform (Schwarz and Zehr 2021). The participants completed the experiment remotely on their own computers via a link distributed through Prolific. The experimental sentences were presented in a box, and the participants clicked "continue" to proceed to the question. The question was in the format

of “What is ‘one/ones’ referring to?” for the Anaphoric *one* sentences, and “What is missing after [the licenser]?” for the NPE sentences. The participants had two answer options to choose from. Upon entering the experiment, the participants were presented with informed consent and instructions. They were required to answer every question to proceed to the next question. The participants were instructed to read the sentences at a natural pace and respond to the questions as accurately as possible. The experiment lasted approximately 15 minutes. An example of an Anaphoric *one* experiment presented to human participants can be found in Appendix B.

The accuracy of the human responses was calculated by determining the number of correct answers selected between the two options.

4 Results

Table 9 shows the performance of all the models on the different datasets in the forced-choice experiment.

Table 9

Accuracy of antecedent resolution from BERT, RoBERTa, zero-shot LLMs, and humans. Human accuracy was calculated by first averaging participants’ responses for each item and then computing the overall mean. The best performance for each dataset is highlighted in bold. See Appendix C for a graphical summary.

Test data	BERT	RoBERTa	LLaMA	Gemma	GPT	Human
GAP test	58.60%	76.06%	63.35%	81.55%	83.50%	
WSC 273	62.00%	81.75%	86.08%	82.05%	74.73%	
GAO corpus	78.18%	83.49%	71.48%	85.91%	90.60%	90.69%
DFAO	67.00%	79.22%	61.84%	79.61%	81.58%	88.98%
NFAO	67.30%	79.06%	57.04%	78.17%	80.99%	88.44%
GNPE corpus	73.14%	83.56%	62.16%	89.19%	93.92%	88.01%
DFNPE	66.87%	80.16%	54.37%	81.88%	88.12%	86.35%

4.1 Language Model Performance

As benchmark datasets of the coreference resolution, we tested the model performance on the GAP (Webster et al. 2018) and WSC 273 (Levesque, Davis, and Morgenstern 2012) datasets, which mainly contained common anaphora such as pronouns. Overall, zero-shot LLMs outperformed BERT and RoBERTa. While this finding appears to contrast with prior work where BERT-based models were reported to outperform GPT (Ćavar et al. 2024), this difference may be partly attributable to variations in model versions and task demands.

In most datasets, the LLMs consistently outperformed BERT and RoBERTa, with the exception of LLaMA, which performed worse than RoBERTa. To maintain comparability with human results, where humans were not explicitly trained on pronoun resolution using GAP, we focused on zero-shot LLM results in further analyses. Detailed results from the fine-tuned models are provided in Appendix C.3. Notably, zero-shot LLMs outperformed BERT and RoBERTa, suggesting that LLMs are more capable of handling tasks without direct exposure to finely-curated labeled data. However, since our corpora were sourced from Wikipedia, and all the language models were trained on large datasets that likely included Wikipedia content, there remains a potential issue of data leakage. In that case, LLMs’ superior performance over BERT and RoBERTa may instead reflect more effective exploitation of web-based training data.

4.2 Comparing LLMs and Humans

We conducted statistical tests to compare the performance of the different LLMs and humans. Specifically, we used McNemar’s test (McNemar 1947) for comparisons among LLMs, as the data involved paired binary samples. For comparisons between LLMs and humans, we employed the Wilcoxon signed-rank test (Wilcoxon 1945), which accommodates both paired binary and continuous data. The complete results are shown in Table 10. In the GAO corpus, GPT-4o mini outperformed the other LLMs and achieved human-level performance (GPT-4o mini: 90.6% vs. Humans: 90.69%, $p = 0.91$). For the NPE resolution tasks, in the GNPE corpus, GPT-4o mini outperformed both other LLMs and humans (GPT-4o mini: 93.92% vs. Humans: 88.01%, $p < 0.001$). These results indicate that, for both Anaphoric *one* and NPE resolution tasks, the best-performing LLM, GPT-4o mini, achieved human-level or superior performances.

Table 10

Statistics for comparisons among humans and models. Comparisons between the human results and models were conducted using the Wilcoxon signed-rank test, with the statistics W reported. Comparisons between models were conducted using McNemar’s test, with the statistics χ^2 reported. For all comparisons, associated p -values are reported in parentheses.

Dataset	Human (W)			GPT (χ^2)		Gemma (χ^2)
	GPT	Gemma	LLaMA	Gemma	LLaMA	LLaMA
GAO corpus	1,331.0 (0.910)	1,328.0 (0.018)	1,189.0 ($< .001$)	5.63 (0.018)	33.72 ($< .001$)	18.57 ($< .001$)
DFAO	461.0 (0.014)	511.0 (0.007)	329.5 ($< .001$)	0.21 (0.65)	13.14 ($< .001$)	10.09 (0.002)
NFAO	428.5 (0.014)	504.0 ($< .001$)	375.0 (0.026)	0.45 (0.50)	18.15 (0.005)	14.02 ($< .001$)
GNPE corpus	1297.0 ($< .001$)	2570.0 (0.400)	1795.0 ($< .001$)	6.04 (0.014)	83.16 ($< .001$)	58.88 ($< .001$)
DFNPE	1075.5 (0.55)	1045.5 (0.180)	637.5 ($< .001$)	4.05 (0.044)	40.13 ($< .001$)	27.19 ($< .001$)

4.3 Influence of Distance and Morphological Information

The distance factor reduced performance for both humans and zero-shot LLMs, though the effect was smaller for humans. The complete statistics results are shown in Table 11. For the Anaphoric *one*, human performance dropped slightly from 90.69% (GAO) to 88.98% (DFAO), while the best-performing LLM, GPT-4o mini, dropped from human-level performance of 90.6% (GAO) to a significantly worse performance of 81.58% (DFAO). A similar pattern was observed for NPE, where human performance decreased only modestly from 88.01% (GNPE) to 86.35% (DFNPE), whereas LLMs’ performance had a visible decrease (e.g., GPT-4o mini performed 88.12% in DFNPE vs. 93.92% in GNPE). The number factor exhibited a similar pattern, with the performance hindered to a degree comparable to that of the distance factor. BERT and RoBERTa also showed decreased performance in the distance and number factor sub-corpora. These findings suggest that both encoder and decoder models were sensitive to distributional factors such as increased distance and number mismatches (Devlin et al. 2019; Liu et al. 2019).⁴

⁴ In the original set of experiments, we tested BERT and RoBERTa on 300-item subsets from the GAO and GNPE corpora. Experiment I tested the GAO and GNPE corpora without fine-tuning BERT and RoBERTa on these datasets, while Experiment II fine-tuned BERT and RoBERTa on the first 210 items and tested the models on the remaining 90 items of the 300-item subsets from the GAO and GNPE corpora. We conducted a third experiment to test model performance on the distance and number sub-corpora. In this

Table 11

Statistics for comparisons between datasets with different linguistic factors (distance and number) for each model. We used proportional tests and reported the z statistics. For all tests, associated p -values are reported in parentheses.

LLMs/Human	GAO		GNPE
	DFAO	NFAO	DFNPE
LLaMA	2.08(0.038)	3.01(0.003)	1.62(0.106)
Gemma	1.71(0.086)	2.04(0.041)	2.19(0.029)
GPT	2.74(0.006)	2.85(0.004)	2.16(0.031)
Human	0.62(0.530)	0.85(0.400)	0.54(0.590)

When comparing the two types of anaphoric elements, NPE was more difficult than Anaphoric *one* for humans (88.01% vs. 90.69%), whereas two out of the three LLMs (GPT-4o mini and Gemma) showed the opposite pattern, achieving higher accuracy on NPE (e.g., GPT-4o mini: NPE 93.92% vs. *one* 90.6%). Logistic regression indicated that the interaction between the intelligence agent (GPT-4o mini vs. human) and the type of anaphoric element (Anaphoric *one* vs. NPE) was marginally significant ($z = 1.69$, $p = 0.091$).

4.4 Exploratory Error Analysis

Moving beyond overall accuracy rates and sub-corpora comparisons, we conducted further analyses to better understand how humans and language models resolve anaphoric dependencies. We focused on how resolution strategies previously hypothesized for humans influence errors produced by humans and LLMs (Hu et al. 2022, 2023; Hu and Frank 2024). Specifically, we investigated (1) the extent to which errors made by humans and LLMs could be attributed to surface-level statistical cues by conducting a SHAP analysis, and (2) how LLMs and humans' performance varied as a function of syntactic parallelism by categorizing sentences according to the anaphora-resolution strategies they afford and comparing performance across these subcategories.

4.4.1 Influence of Surface-level Statistical Cues: SHAP-based Analysis

Methods. To investigate which surface-level statistical cues influenced errors in antecedent resolution for LLMs and humans, we modeled the relationship between cue features and binary outcomes separately for each LLM and for humans. We then applied SHapley Additive exPlanations (SHAP) to interpret the fitted models. SHAP provides a game-theoretic framework for attributing the contribution of each feature to individual predictions (Lundberg and Lee 2017; Lundberg et al. 2020). By comparing feature attributions across LLMs and humans, we aim to characterize the extent to which LLMs and humans rely on surface-level statistical cues.

For simplicity and interpretability, we defined three features: (1) **Word Frequency** as an approximate of shallow semantic prominence, was calculated as $p_{\text{antecedent}} - p_{\text{alternative}}$, where p denotes the log10 probability assigned by a 5-gram language model trained on Wikipedia corpus computed using KenLM (Heafield 2011; Ponferrada 2021). Higher values indicate that the correct antecedent has a higher frequency than the alternative. (2) **Distance**, which aimed to

experiment, we also tested BERT and RoBERTa on the full datasets (457 GAO items and 391 GNPE items), where BERT and RoBERTa achieved higher accuracies: 84.5% and 86.14% on GAO, and 79.1% and 86.09% on GNPE. However, their performance dropped in the more challenging sub-corpora: 67.0% and 79.22% on DFAO, 67.3% and 79.06% on NFAO, and 66.87% and 80.16% on DFNPE. To avoid confusion, we report the results of BERT and RoBERTa on the GAO and GNPE corpora from Experiment I for consistency across model evaluations, as all LLMs (i.e., LLaMA, Gemma, and GPT) and human participants were evaluated on the same 300-item subsets. Full results from all experiments are available on GitHub (<https://github.com/nayoun1/Anaphoric-One-and-NPE>).

capture the effect of distance, was defined as $d_{\text{antecedent}} - d_{\text{alternative}}$, where d is the distance in terms of number of characters from the candidate antecedent to the anaphor (*one/ones* in GAO, or the licenser in GNPE). Higher values indicate that the correct antecedent is further away from the anaphor than the alternative. (3) **Number**, for Anaphoric *one*, this feature aimed to capture whether the anaphor (*one/ones*) agreed in number with the correct antecedent (1 = agreement, 0 = disagreement). The features showed weak pairwise correlations (with the highest absolute correlation being 0.12) for both *one* and NPE.

We fitted an XGBoost decision tree model separately for each LLM and for humans, as tree-based methods can automatically capture nonlinearities and higher-order feature interactions. SHAP values were then computed from the fitted models to assess the contribution of each feature. Note that the SHAP values reflected the relative rather than absolute contribution of these factors included in the model, as the features included should not be considered as an exhaustive set of all possible surface-level cues.

Results. Tree-based models achieved reasonable performance in predicting whether LLMs and humans answered correctly, as evidenced by reasonable F1 scores using 3-fold cross-validation (LLaMA: $Mean = 0.76$, $SD = 0.04$; Gemma: $Mean = 0.91$, $SD = 0.003$; GPT: $Mean = 0.94$, $SD = 0.01$; Humans: $Mean = 0.96$, $SD = 0.003$). While non-surface cues were not directly measured, the high performance of the models suggests that the effects of other cues may be partially captured indirectly through correlations with the included features.

Figure 3 (left panels) shows the feature contributions derived from the XGBoost model predicting LLMs' and humans' errors for the GAO corpus using SHAP analysis. The upper-left panels of Figure 3 display global feature contributions, where the x-axis is the mean absolute SHAP values across all items. Across all three LLMs, Distance was the strongest predictor of correctness, while for humans Word Frequency was the most influential feature. The lower-left panels of Figure 3 show item-level SHAP value distributions, colored by feature values. Although there was a limited degree of linear relationship between the contributions of these features and their values, LLMs and humans seemed to show different patterns. For Gemma and GPT, lower Word Frequency feature value (i.e. the correct antecedent had much lower word frequency compared to the alternative) yielded positive contributions to correct resolution. In contrast, for humans, lower word frequency of the correct antecedent biased participants toward the incorrect alternative, albeit with weaker effect. In other words, unlike LLMs, humans were more likely to choose the more frequent antecedent. Moreover, LLMs benefited from lower Distance (i.e. when the antecedent is closer to the anaphoric element), whereas for humans, the effect of Distance appeared more nonlinear.

Similarly, Figure 3 (right panels) shows the results for the GNPE corpus. Global feature contributions (upper-right panels) suggested that Distance was the dominant feature for all LLMs and humans except LLaMA. Item-level distributions (lower-right panels) suggested that for Gemma and GPT, SHAP distributions revealed a relatively linear relationship, where lower values of Distance (i.e. when the antecedent is closer to the anaphora) were associated with correct antecedent resolution. For humans, the effect of Distance was not as obvious or linear, but a positive association between Word Frequency and correct responses can be observed, whereas LLMs exhibited weaker or slightly reversed patterns.

4.4.2 Influence of Anaphora Resolution Strategies: Error Rates as a Function of Syntactic Parallelism

Methods. To examine how humans and LLMs use syntactic parallelism during anaphora resolution, we subcategorized sentences according to whether resolution could be achieved by using a syntactic parallelism strategy. The syntactic parallelism strategies considered in this study included Subset (Pearl 2007; Goldberg and Michaelis 2017), Contrasting (LuperFoy 1991; Goldberg and Michaelis 2017; Carlson and Harris 2018), Clausal or VP Coordination (Carlson 2020), Modifier Coordination (LaCara 2018), and Repetition (Goldberg and Michaelis 2017; Carlson 2020); their definitions and examples are provided in Table 12. Each sentence was then annotated based on whether the anaphoric expression (Anaphoric *one* or NPE) exhibited one or more of the parallel relationships listed in the table with its antecedent. This annotation scheme allowed us to distinguish between "parallel" items, in which the anaphoric resolution could be facilitated by any of the syntactic parallelism strategies in Table 12, and "non-parallel" items that lacked these

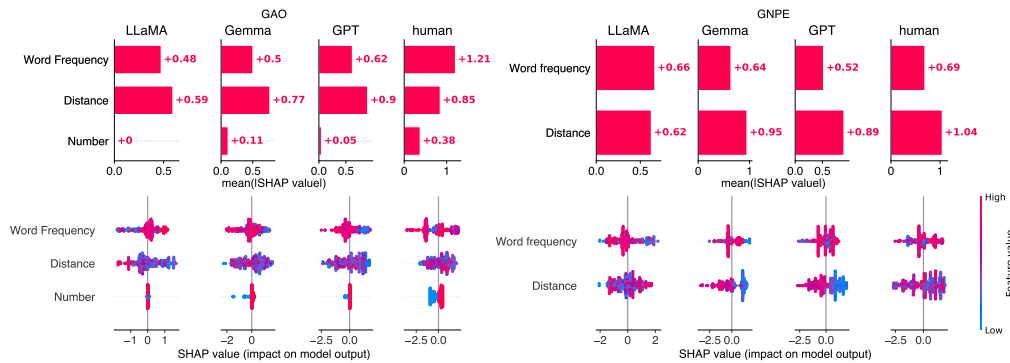


Figure 3
Global (top) and item-level (bottom) SHAP contributions of each feature for LLMs and humans in the GAO (left) and GNPE (right) corpus.

Table 12
Types of Parallelism Strategies for Anaphoric *one* and NPE, with Definitions and Examples.

Type of parallelism	Definition & Example	Portion (<i>one</i> /NPE)
Subset	Anaphoric <i>one</i> /NPE represent a subset of the antecedent. <i>Example:</i> "This trip proves to be a busy <i>one</i> ." / "The river has the steepest fall of <i>any</i> in Wales and England."	24% / 39%
Contrasting	Anaphoric <i>one</i> /NPE appear in a (semantically) contrasting construction. <i>Example:</i> "Not my favorite reading, but a good <i>one</i> ." / "A sentence has a meaning according to A, whereas B can't discern <i>any</i> ."	12% / 16%
Clausal or VP Coordination	Anaphoric <i>one</i> /NPE appear in coordinated clauses or VPs that are parallel to those containing the antecedent. <i>Example:</i> "He didn't see a job I wanted [in Austin] – so I created <i>one</i> ." / "There's not much time, but there is <i>some</i> !"	3% / 26%
Modifier Coordination	Anaphoric <i>one</i> /NPE have the same or contrasting modifier (including adjectives, PPs, or relative clauses) as the antecedent. <i>Example:</i> "Make him a new blanket to replace his old <i>one</i> ."	7% / -
Repetition	Anaphoric <i>one</i> /NPE appear in the same construction as the antecedent. <i>Example:</i> "These will be the last elections! The last <i>ones</i> !" / "She spent "some" of her childhood in Spain and " <i>some</i> " in Massachusetts."	1% / 1%
non-parallelism	None of the strategies apply.	53% / 18%

cues and required reliance on more complex syntactic, semantic, and referential information in the discourse to identify an antecedent.

Consistent with previous theoretical work, Anaphoric *one* is primarily resolved by linking it to referential representations within a discourse model (Culicover and Jackendoff 2005; Payne, Derbyshire, and Pullum 2013; Goldberg and Michaelis 2017) and is less likely to rely on parallelism-based strategies. This pattern is reflected in the item distribution: only 47% of

Anaphoric *one* items involved potential parallelism strategies, while 53% did not. In contrast, parallelism played a much larger role in NPE resolution, with 82% of NPE items involving potential parallelism strategies and only 18% lacking such strategies.

Results. Table 13 reports error rates for LLMs and humans as a function of whether sentences could be resolved using a parallelism strategy. Overall, both LLMs and humans showed lower error rates on items involving parallelism (collapsing across all parallelism types) than on non-parallelism items, with the exception of LLaMA in the NPE forced-choice task, where error rates were highest across corpora and strategies. Notably, LLMs showed a greater benefit from parallelism than humans: while human accuracy increased by less than 5%, accuracy gains for Gemma and GPT were at least twice as large. Exploratory statistical analyses revealed that the effect of parallelism reached significance for Gemma and GPT, but not for humans or LLaMA for either *one* and NPE (see Table E.1 in Appendix E). In addition to parallelism, we also examined the role of semantic plausibility through a plausibility rating experiment (see Appendix E for full details).

Table 13

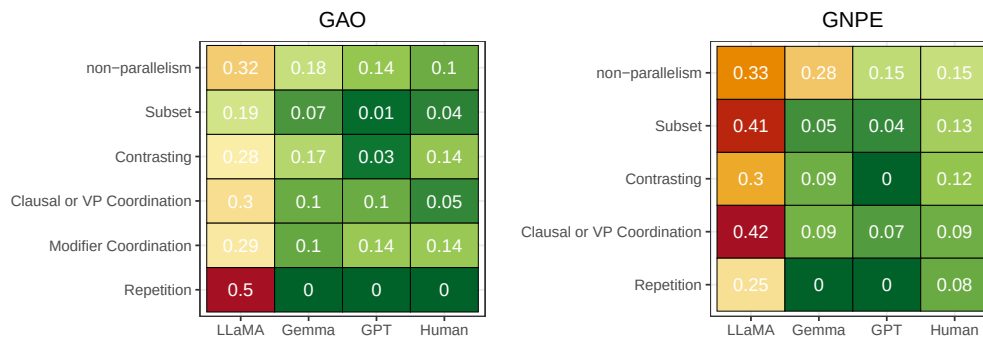
Error rates of antecedent resolution from zero-shot LLMs and humans in the forced-choice task, split by parallelism strategies.

Dataset	Strategy	LLaMA	Gemma	GPT	Human
GAO	non-parallel	32.28%	18.35%	13.92%	10.18%
	parallel	24.11%	9.93%	4.26%	8.27%
	Difference	8.17%	8.42%	9.66%	1.91%
GNPE	non-parallel	33.33%	27.78%	14.81%	15.43%
	parallel	38.84%	7.02%	4.13%	11.23%
	Difference	-5.51%	20.76%	10.68%	4.20%

Figure 4 presents error rates for each parallelism strategy. For non-parallel sentences, humans' error rates were the lowest across Anaphoric *one* and NPE. In contrast, for parallel sentences, humans' error rates were comparable to or higher than those of Gemma and GPT, depending on the type of anaphora and the LLM compared. For NPE, GPT and Gemma exhibited lower error rates than humans across all parallelism types, especially Subset and Contrasting. As prior findings have mainly highlighted reading-time facilitation effects of various parallel structures in ellipsis resolution (Harris and Carlson 2017; Carlson 2020; Kim et al. 2020b), GPT and Gemma appeared to benefit from Subset and Contrasting structural regularities more effectively than humans. For *one*, GPT showed the lowest error rates on Subset and Contrasting structures, Gemma showed the lowest error rate on Modifier Coordination, while humans showed the lowest error rate on Clausal or VP Coordination. As *one* serves discourse functions such as contrasting salient alternatives or selecting a representative exemplar (LuperFoy 1991; Goldberg and Michaelis 2017) and introducing a context-driven similarity relation (Culicover and Jackendoff 2012; Goldberg and Michaelis 2017), GPT appeared to be especially facilitated by Subset and Contrasting structures. Overall, GPT and Gemma consistently leveraged parallelism cues, showing broad sensitivity in NPE and increased sensitivity to Subset and Contrasting relations for Anaphoric *one*. For coordination structures (Clausal or VP Coordination and Modifier Coordination for *one*), humans were almost as effective as LLMs in leveraging such parallelism information. Although items with Repetition-based parallelism were rare, both humans and LLMs achieved the highest performance and resolved the anaphora almost always successfully.

4.5 Open-ended Antecedent Identification Experiment

Methods. In the forced-choice experiment, we provided two candidate antecedents, which may have supplied too much information. We also observed that LLMs sometimes produced "neither" responses (LLaMA: 0%/0%, Gemma: 3.0%/1.3%, GPT: 3.3%/0%, for *one*/NPE respectively), which were automatically scored as "FALSE." To better understand how humans and LLMs

**Figure 4**

Error rates of antecedent resolution from zero-shot LLMs and humans by parallelism strategies in the forced-choice task for GAO (left) and GNPE (right). Cell colors range from green (low error rates) through yellow to red (high error rates), indicating the magnitude of error rates for each parallelism strategy. Parallelism strategies are defined as in Table 12.

employ different strategies to resolve anaphoric dependencies, we therefore conducted an open-ended antecedent identification experiment. As LLMs' performance depends on both anaphoric resolution ability and task demands (Hu and Frank 2024), we expected overall lower performance in the open-ended antecedent identification experiment (i.e., we expect to observe a "demand gap"). Beyond this overall decline, we ask whether the influence of different linguistic properties and recovery mechanisms on model performance persists in the same way as in the forced-choice task, relative to human performance.

In this open-ended antecedent identification experiment, humans and LLMs were asked to generate the antecedent NP of Anaphoric *one* and NPE rather than selecting between two candidates. We used the first 300 sentences from our GAO and GNPE corpora for the Anaphoric *one* and NPE resolution tasks, testing both with LLMs and human participants (N=130). All other aspects of the setup were the same as in the forced-choice experiment. For the error analysis, we annotated and analyzed specific types of errors produced by LLMs and humans.

Results. In the open-ended antecedent identification task, exact string matching between generated and expected answers yielded low accuracy, as many responses differed only superficially from the expected answers. We manually re-evaluated the outputs, applying the following rules: (1) responses with missing determiners (e.g., "people" vs. "the people") or number mismatches (e.g., "song" vs. "songs") were reclassified as correct; (2) responses that differed in form but conveyed the full meaning (e.g., "sick or dying dogs" for "dogs that were sick or dying") were also reclassified as correct; and (3) for complex NP antecedents (antecedent NP consisting of a head noun and one or more modifiers), responses containing only the head noun were retained as incorrect. The same rules were applied to both human and LLM responses. Examples of the re-examination can be found in Appendix D (see Tables D.1 & D.2), and the full re-examination sheet of humans and three models in two anaphoric resolution tasks can be found in the supplementary material on GitHub (<https://github.com/nayoun1/Anaphoric-One-and-NPE>).

The accuracies based on the manually rechecked results are shown in Table 14. Note that task type had a greater impact on LLMs: LLMs' performance dropped substantially in the open-ended identification task relative to the force-choice task (cf. Table 9), whereas human performance remained stable. The decline in LLM performance could be attributed either to overestimated performance in the forced-choice task, which indicated LLMs' lack of human-level capacity in anaphora resolution, or to increased task demands in the open-ended identification task, which should not be taken as evidence of an inherent incapacity in LLMs (see Section 5.3 for discussion). Task framing may lead LLMs to prioritize the most semantically prominent head noun governing the syntactic category of the phrase. Despite this ambiguity in interpretation, with respect to our focus on the effects of linguistic properties on anaphora resolution, the pattern of results was qualitatively consistent with that observed in the forced-choice experiment (Table 9). First,

GPT outperformed the other two LLMs, and LLaMA performed the worst. Second, distance and number factors influenced overall performance, with stronger effects on LLMs than on humans (though distance exerted a greater influence on humans in resolving Anaphoric *one*). Third, NPE resolution was more difficult for humans, whereas Anaphoric *one* was more difficult for Gemma and GPT. Statistical test results are reported in Appendix D (see Tables D.3 & D.4).

Table 14

Manually examined Accuracy rates of GPT, LLaMA, Gemma, and humans on the Anaphoric *one* and NPE datasets (with the same datasets as in Table 9).

	LLaMA	Gemma	GPT	Humans
GAO corpus	22.67%	46.67%	65.67%	90.50%
DFAO	22.22%	45.10%	58.82%	85.19%
NFAO	17.52%	42.34%	56.93%	88.69%
GNPE corpus	8.67%	66.00%	67.33%	86.89%
DFNPE	11.66%	50.92%	55.21%	83.64%

Table 15 presents an analysis of error types based on open-ended responses. During re-examination, we identified several systematic behaviors. First, while both humans and LLMs occasionally adjusted the number of the predicted antecedent to agree with the anaphoric element (i.e., Anaphoric *one/ones*) or its licensors, this morphological alignment was more frequent in LLM outputs. In the NFAO sub-corpus of Anaphoric *one* resolution, errors due to number alignment with the target Anaphoric *one/ones* accounted for 58.8% of all GPT's errors, 47.1% of Gemma's, and 35.3% of LLaMA's. Second, LLMs (especially Gemma and GPT) were more likely than humans to produce "head-only" answers, in which only the head noun of a complex antecedent NP was provided.

Table 15

Proportions of two types of errors ("Number Alignment" & "Head-only" Answers) by three LLMs and Human.

Error Type	Anaphora	LLaMA	Gemma	GPT	Human
"Head-only" Answers	Anaphoric <i>one</i>	2% (4/224)	8.5% (15/177)	37.6% (47/125)	1% (1/84)
	NPE	0%	28% (35/124)	9% (9/103)	4% (5/120)
"Number Alignment"	Anaphoric <i>one</i>	6% (13/224)	7.5% (13/177)	16.8% (21/125)	10% (9/84)
	NPE	1% (4/275)	13% (16/124)	15% (15/103)	7.5% (9/120)
Others	Anaphoric <i>one</i>	92% (207/224)	84% (149/177)	45.6% (57/125)	89% (74/84)
	NPE	99% (271/275)	59% (73/124)	76% (79/103)	88.5% (106/120)

5 Discussion

This study investigated how language models identify antecedents for two types of anaphoric elements—Anaphoric *one* and NPE—which differ in their grammatical properties. Although both depend on the presence of an antecedent, psycholinguistic studies suggest that humans access antecedents through distinct underlying mechanisms. These differences stem from NPE's greater

sensitivity to grammatical information, as noted in prior research (Jackendoff 1977; Lobeck 1991, 1995; Llobart-Huesca 2002; Lobeck 2006; Günther 2013; Kim, Brehm, and Yoshida 2019). By contrast, Anaphoric *one* is less constrained by syntactic licensing and instead relies more on semantic and discourse representations for antecedent access (Hankamer and Sag 1976; Halliday and Hasan 1976; Panagiotidis 2003; Culicover and Jackendoff 2005; Goldberg and Michaelis 2017).

Against this theoretical backdrop, this study addressed two main research questions: (i) To what extent do language models exhibit human-like abilities in retrieving information associated with antecedents and employ similar strategies to resolve Anaphoric *one* and NPE, given their distinct linguistic properties and the mechanisms involved in antecedent recovery? (ii) Are language models influenced by surface linguistic features (distance and morphological features), parallelism, and more complex cues to the same degree as humans?

By comparing the performance of various language models and humans in resolving these two types of anaphoric elements and examining the effects of linguistic factors that influence the identification of the correct antecedent for *one* and NPE, this study provides a starting point for an overall evaluation of language models' abilities to resolve complex anaphoric structures.

5.1 Summary of Results

First, we introduce a new dataset drawn from naturalistic web corpora targeting two complex anaphoric constructions in English: Anaphoric *one* and NPE. Linguistically, these constructions are particularly interesting because both depend on antecedents for interpretation, but in different ways: *one* requires access to semantic and referential information in discourse (Culicover and Jackendoff 2005; Payne, Derbyshire, and Pullum 2013; Goldberg and Michaelis 2017), whereas NPE primarily relies on structural cues (Kim, Brehm, and Yoshida 2019; Kim et al. 2020a). Previous research shows that humans experience processing difficulty with both types of anaphora, though they rely on different information sources—typically structural information for NPE and semantic and referential discourse information for *one*. We provide detailed annotations with regard to potential distractors in the context, distance and number agreement, and parallelism strategies, which offer superficial morphological and syntactic cues that potentially contribute to anaphora resolution. To our knowledge, this is the first dataset to jointly examine Anaphoric *one* and NPE, enabling rich possibilities for exploring anaphora resolution mechanisms for LLMs and humans from multiple perspectives.

Second, we examined human and LLM performance in a forced-choice task. Overall, GPT-4o significantly outperformed the other tested LLMs, and its performance was comparable to or exceeded that of humans (Table 9 & 10). Distance and morphological information influenced both LLMs and humans' performance: increased distance between the anaphora and the antecedent led to reduced performance for both *one* and NPE, and number mismatch between the anaphora and the antecedent (which was only relevant for *one*) also led to reduced performance. This pattern was consistent with previous psycholinguistic findings (Kim, Brehm, and Yoshida 2019; Kim et al. 2020a,b). However, LLMs' performance was more strongly influenced by distance and morphological information, whereas humans were more robust (Table 9 & 11).

Third, we further examined the influence of surface statistical cues and syntactic parallelism strategies on models' performance through exploratory analyses and an open-ended antecedent identification experiment. Specifically, distance better predicted LLMs' errors and word frequency better predicted humans' errors in resolving *one*, whereas distance better predicted errors in resolving NPE (Figure 3). This pattern is consistent with the interpretation that LLMs are sensitive to surface-level cues, while humans rely more on syntactic, semantic, and referential information in discourse. Analyses of syntactic parallelism strategies showed that LLM performance increased to a greater extent when a syntactic parallelism strategy was available, whereas humans benefited less from such strategies (Table 13), again consistent with the interpretation that LLMs rely on relatively simple syntactic strategies while humans integrate syntactic, semantic, and referential information more flexibly. Results from the open-ended identification experiment (Table 14) further showed that the best-performing LLM, GPT-4o, performed worse on sub-corpora with structurally more difficult items, while human performance remained stable. This pattern replicates the forced-choice task results (Table 9). Although overall performance declined in the open-ended task, this decrease likely reflects limitations in LLMs' task interpretation or response generation, rather than deficits in underlying linguistic knowledge.

5.2 The Importance of These Corpora for Linguistic, Psycholinguistic, and Computational Research

The two linguistically distinct anaphoric elements (Anaphoric *one* and NPE) pose different challenges for anaphora resolution. NPE resolution requires reconstructing the semantic and syntactic representations of the antecedent, a process that is challenging for humans because it demands identifying, accessing, and integrating syntactic, semantic, and discourse information about the antecedent (Martin and McElree 2008, 2009, 2011; Kim, Brehm, and Yoshida 2019). By comparison, Anaphoric *one* resolution relies less on syntactic constraints and more heavily on accessing semantic and referential information within the discourse (Jackendoff 1977; Culicover and Jackendoff 2005; Payne, Derbyshire, and Pullum 2013; Goldberg and Michaelis 2017).

Humans generally resolve anaphora by integrating syntactic, semantic, and discourse-level information (Gordon, Grosz, and Gilliom 1993; Tanenhaus et al. 1995; Ferreira, Bailey, and Ferraro 2002; Ferreira and Patson 2007; Kehler et al. 2008; Jäger, Engelmann, and Vasishth 2017). By contrast, LLMs appear to rely more heavily on surface-level cues such as syntactic parallelism, linear proximity, and morphological agreement. While humans may occasionally misidentify antecedents when discourse-level plausibility outweighs structural constraints (Van Dyke and McElree 2006; Van Dyke 2007), LLMs often perform comparatively well on NPE by exploiting simpler structural patterns and a limited set of licensors. Although these potential mechanisms are neither mutually exclusive nor exhaustive, and intelligent systems may use them in combination, our corpora provide a useful framework for future work in distinguishing between surface-driven and semantically and referentially driven dependencies in discourse.

In addition to the two AO-NPE datasets we created from English Wikipedia texts with rich linguistic annotations (e.g., licensor type, distance, number agreement, and parallelism), we also constructed simplified AO-NPE datasets (see Weissweiler et al. 2025 for a discussion of ideal benchmarks capturing flexible and gradient generalization in human language) extracted from the enTenTen21 corpus via Sketch Engine (Kilgariff et al. 2014; see Appendix F).

5.3 LLMs' Capabilities and Limitations in Resolving Complex Anaphora

Overall, our findings revealed that although language models demonstrated the capacity to resolve Anaphoric *one* and NPE, their performance appeared less robust than that of humans when facing challenges in referential dependencies and task framing. These findings align with previous studies on language models' capabilities in anaphora resolution (Rambelli et al. 2024; Warstadt et al. 2020). Although neural models using contextualized representations have substantially improved compared to earlier models, they still face challenges in syntactic and semantic domains and generally perform worse than humans (Khullar, Majumdar, and Shrivastava 2020; Warstadt et al. 2020; Hardt 2023; Ćavar et al. 2024; Kobzeva et al. 2025).

The performance of language models depends on several factors, including model architecture, training methods, and the task design (Liu, González Pellicer, and Gillick 2016). Although we observed overall reduced accuracy for LLMs in the open-ended identification task compared to the forced-choice task, we could not conclude that LLMs lack human-like linguistic knowledge or rule out the possibility that task framing plays a role. Hu and Levy (2023) show that LLMs' performance as measured by metalinguistic judgments is inferior to performance based on direct measurements of probability distributions, which can lead to an underestimation of LLMs' linguistic knowledge. They further suggest that constructing minimal pairs allows fair comparison between different intelligent systems. We consider this more rigorous approach a promising direction for future work on assessing LLMs' anaphora resolution abilities, particularly through corpora designed around targeted hypotheses, informed by our exploratory findings on complex anaphora resolution.

Another example highlighting the importance of task design is Ćavar et al. (2024). They focus on ellipsis detection and localization, which can be framed as a structured sequence labeling problem. Such tasks benefit from supervised fine-tuning, a setting in which encoder-based models like BERT typically excel. In contrast, our study examines antecedent identification for Anaphoric *one* and NPE in a forced-choice setting. This task places greater emphasis on comparing candidate antecedents rather than reconstructing missing structure, making it inherently less aligned with BERT's pre-training approach based on a masked language modeling objective (i.e., fill-in-the-

blank). These differences between the tasks may explain why BERT performs well in Čavar et al. (2024) but not as well in our setting.

One observation worth mentioning concerns the specificity of LLMs' responses. In the open-ended identification task, LLMs often produced "head-noun-only" responses (e.g., "dogs" for "dogs that were sick or dying"), suggesting a tendency to default to core semantic content and category information rather than recovering full structural information. This pattern is consistent with prior work showing that NPE resolution typically involves recovering a syntactically parallel structure to the antecedent, and that identifying only the head noun or its features may not capture the full representation required (Kim, Brehm, and Yoshida 2019; Kim et al. 2020a). This pattern also aligns with prior findings that LLM performance declines when identifying exact mention boundaries, compared to identifying only the head (Novák et al. 2025), indicating that reconstructing full referential spans remains challenging. While this limitation may partly reflect task design or prompting effects, it points to a broader difficulty in capturing complete structural representations (McCoy, Frank, and Linzen 2020; Linzen and Baroni 2021; Marvin and Linzen 2018; Petty and Frank 2021; Hahn 2020; Warstadt et al. 2020). This issue is relevant for downstream tasks such as summarization and translation, which benefit from explicit anaphora resolution during preprocessing (Lo and Mitkov 2025), highlighting the need for accurate antecedent reconstruction.

5.4 Mechanisms of Anaphoric Resolution for LLMs and Humans

In response to Research Question 2, our results revealed that although both humans and LLMs were challenged by surface-level factors such as parallelism, linear distance, and number agreement between the anaphoric elements and their antecedents, LLMs were more strongly influenced by these factors, compared to humans (Linzen, Dupoux, and Goldberg 2016; Emami et al. 2019; Sukthankar et al. 2020; Warstadt et al. 2020). Quantitatively, LLMs showed a larger performance drop in accuracy in both the forced-choice task and the open-ended identification task, whereas human performance remained relatively stable (Tables 9, 13, 14). Qualitatively, LLMs often adjusted predicted antecedents to match the number of the anaphoric element (i.e., Anaphoric *one/ones*) or licensor (e.g., predicting "songs" after "two") in the open-ended identification experiment, a pattern less common in human responses, and one that reflects LLMs' tendency to rely on morphological information such as agreement (Table 15).

While language models often display human-like sensitivity (Golan et al. 2023; Amouyal, Meltzer-Asscher, and Berant 2024), it remains unclear whether this arises from genuine structural linguistic knowledge, as it does in humans. Humans are known to combine rich morphological, syntactic, and semantic cues in anaphoric resolution (Jäger, Engelmann, and Vasishth 2017) and adjust their reliance on different cues depending on linguistic properties (Mertzen et al. 2023), weighting syntactic and semantic/discourse influences differently. Predictions and model probability distributions underlying these expectations may also differ from those of humans. While Huang et al. (2024) suggest that next-word prediction from language models is not the only mechanism that governs processing difficulty, other works clearly note its predictive role in language comprehension, though the form and function likely differ between current LLM training and humans (DeLong, Urbach, and Kutas 2005; Federmeier 2007; Kutas, DeLong, and Smith 2011; Bender and Koller 2020; Hu and Levy 2023; Cuskley, Woods, and Flaherty 2024; Hu and Frank 2024; Shain et al. 2024). This discrepancy may lead to different sensitivities to the impact of linguistic factors on language models' and humans' performance in anaphoric resolution, with humans generally demonstrating greater robustness to these linguistic factors than models.

5.5 Limitations and Future Directions

A further consideration is why some models occasionally outperformed humans on NPE resolution despite its structural complexity. For language models, identifying the antecedent for NPE may not be particularly challenging, as the sentences in our corpora are generally clear and well-formed, with ellipsis typically used only when the intended meaning is easily recoverable. Furthermore, the set of licensors used (e.g., *any*, *many*, etc.) is relatively limited (Lobeck 1995; Günther 2013) and likely occurred frequently in the models' pre-training data. Thus, even though NPE involves deeper syntactic licensing constraints (Hankamer and Sag 1976; Sag and Hankamer

1984), LLMs can often resolve these dependencies by relying on predictable antecedent patterns. Future research should examine a wider range of NPE constructions across diverse syntactic environments, particularly those in which antecedent recovery depends on more complex structural relationships (Kush 2013; Kobzeva et al. 2025).

Although this interpretation should be approached with caution, the detailed grammatical and semantic information associated with anaphoric elements, which differentiates Anaphoric *one* and NPE, may not be easily encoded in language models (Linzen, Dupoux, and Goldberg 2016; Kuncoro et al. 2018, 2019; Hahn 2020; McCoy, Frank, and Linzen 2020; Petty and Frank 2021; Wilson, Zhou, and Frank 2023). These results suggest that models are sensitive to distributional and morphological content, but exhibit certain limitations in capturing the underlying syntactic, semantic, and morphological principles that govern ellipsis and Anaphoric *one* (Khullar, Majmundar, and Shrivastava 2020; Hardt 2023; Testa, Chersoni, and Lenci 2023; Ćavar et al. 2024).

Finally, the present study focused on the final outputs produced by both LLMs and humans, rather than on the incremental online processes that underlie these responses. Recent work has begun to investigate the internal mechanics of LLMs during real-time processing by comparing layer-wise metrics to human processing measures, such as reading times. These studies suggest that LLMs can be examined not merely as black-box input-output systems but also as potential computational models of language processing (Wilcox, Vani, and Levy 2021; Van Schijndel and Linzen 2021; Arehalli, Dillon, and Linzen 2022; Huang et al. 2024; Kobzeva and Kush 2024; Hu, Lepori, and Franke 2025). This line of research raises broader questions about whether LLMs employ strategies and heuristics comparable to those used by humans and how these unfold over time. Future research should aim to draw more direct comparisons between human and model performance during real-time processing, and examine whether patterns of model errors mirror those made by humans in real-time processing (Arehalli and Linzen 2024; Huang et al. 2024).

6 Conclusion

This study examined the performance of language models in identifying the correct antecedents for anaphoric elements with distinct grammatical properties. To this end, we constructed a novel dataset examining anaphoric *one* and NPE, enabling multifaceted investigations of anaphora resolution in humans and LLMs. Our findings revealed that model performance was influenced by details of the training data, the linguistic properties associated with the anaphora (Culicover and Jackendoff 2005; Lobeck 1995; Merchant 2001; Llobart-Huesca 2002; Payne, Derbyshire, and Pullum 2013; Goldberg and Michaelis 2017; Kim, Brehm, and Yoshida 2019), and the level of model advancement. Our results show that more capable LLMs, such as GPT-4o mini, achieved antecedent accuracy comparable to humans, whereas less advanced models such as LLaMA underperformed. Through further analyses of statistical cues and resolution strategies previously hypothesized for humans, we observed that humans and LLMs exhibited partially divergent difficulty profiles: while both human and LLM anaphoric-resolution performance depended on surface-level statistical cues (e.g. linear distance and number agreement) as well as surface regularities such as parallelism, LLMs showed especially large drops on more complex sentences (e.g., those involving distance or number mismatches and those not resolvable via parallelism), whereas human performance remained relatively stable. This pattern was observed across both forced-choice and open-ended tasks. These findings are consistent with the view that humans often employ syntactic, semantic, and discourse information flexibly, whereas LLMs appear to rely more on surface regularities such as parallelism, frequent licensors, and surface-level factors such as linear distance and number agreement (Gordon, Grosz, and Gilliom 1993; Kehler et al. 2008; Warstadt et al. 2020). At the same time, our results highlight the importance of design decisions such as dataset properties and task demands in shaping observed performance. Together, these findings motivate further investigation into how humans and LLMs resolve complex anaphora and how different task framings influence this process.

Appendix A Further Examples and Annotation Explanation of the AO-NPE Datasets

1. The A(naphoric)O(ne) Corpora

1.1 The G(eneral)AO Corpus

Table A.1

Examples from the GAO corpus

column 1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19
text ID	text	one/ones	antecedent type	one-start character offset	one-end character offset	A	A-start character	A-end character	B	antecedent start character offset	B-antecedent offset	A-antecedent	B-antecedent	A-antecedent	B-antecedent	antecedent number	one-licensor category	URL
1	For those who manipulate the story, may your careers be extremely difficult until you tell the truth... This is not about your ratings, this is about a tragic loss... The citizens are not arming themselves against the police, there is no war between us and you cannot create one!	one	linguistic	275	277	war	238	240	police	218	223	TRUE	FALSE	local	non-local	npsg	verb	https://en.wikipedia.org/wiki/2009_shootings_of_Oakland_police_officers
2	Your actions let us know that they that these officers, did not die in vain. That the people that they were there to serve, were the ones that helped them in the end. The "brave man" he was thanking was Clarence Ellis who had used his own coat to tamponade the arterial blood spurting from Sergeant Mark Dunakin's neck.	ones	linguistic	135	138	officers	47	54	people	87	92	FALSE	TRUE	non-local	local	nppl	det	https://en.wikipedia.org/wiki/2009_shootings_of_Oakland_police_officers
3	Wilson later said that Knott "was a friend of mine who wasn't a songwriter but he contributed a couple of lines. But I can't remember which ones!"	ones	discourse	140	143	songwriter	64	73	lines	106	110	FALSE	TRUE	non-local	local	nppl	det	https://en.wikipedia.org/wiki/Add_Some_Music_to_Your_
4	Grunge was in and, you know what, that's OK, 'cause music needs to do that. It needs to change all the time. That's how it stays fresh. Alias is still Steve and I. Alias will always be Steve and I. Steve and I will do a record tomorrow if the fans need one! Of the same recording the review in American Record Guide was less effusive: "Not my favorite reading, but a good one!"	one	linguistic	253	255	record	220	225	fans	243	246	TRUE	FALSE	non-local	local	npsg	verb	https://en.wikipedia.org/wiki/Alias_(band)
5	Mike D. Brownell criticized Cooke's playing for allmusic.com, claiming that Cooke "gets carried away musically..."	one	linguistic	115	117	review	27	32	reading	95	101	FALSE	TRUE	non-local	local	npsg	adj	https://en.wikipedia.org/wiki/Antony_Cooke

Note. All three AO corpora share an identical annotation format, in which a total of 19 columns are included:

- **Column 1:** Marks the text number. "1" means that the text is the first example in the GAO corpus. The above example lists the first 5 texts in the GAO corpus.
- **Column 2:** Presents the raw text collected from English Wikipedia. The word in bold represents the Anaphoric *one* (*ones*), and the underlined word represents the correct antecedent of the Anaphoric *one*.
- **Column 3:** Marks the Anaphoric *one*. Depending on the number feature, it can be either *one* or *ones*. For example, in texts **ID 1, 4, and 5**, the Anaphoric *one* carries the [**+singular**] number feature, as indicated by the "**one**" in the column. In contrast, in texts **ID 2 and 3**, the Anaphoric *one* carries the [**+plural**] number feature, as shown by the "**ones**" in the column.
- **Column 4:** Marks the type of the antecedent (either linguistic or discourse; (Hankamer and Sag 1976; Sag and Hankamer 1984)) of the Anaphoric *one*. "**Linguistic**" means that the antecedent appears within the same sentence as the Anaphoric *one*, as seen in texts **ID 1, 2, 4, and 5**. In these cases, the antecedent is contained within the same sentence. In contrast, "**discourse**" means that the antecedent is found in the preceding context, as seen in text **ID 3**. In this sentence, the antecedent is not within the same sentence as the Anaphoric *one*.
- **Columns 5 & 6:** Marks the start and end character offsets of the Anaphoric *one*.
- **Column 7:** Marks the first antecedent candidate, which may be either the correct or incorrect antecedent of the Anaphoric *one*, as indicated in column 13. For example, in texts **ID 1 and 4**, **antecedent A** is the **correct antecedent** (marked as "**TRUE**" in Column 13) of the Anaphoric *one*, whereas in texts **ID 2, 3, and 5**, **antecedent A** is **not the correct antecedent** (marked as "**FALSE**" in Column 13).
- **Columns 8 & 9:** Marks the start and end character offsets of antecedent A.
- **Column 10:** Marks the second antecedent candidate, which may be either the correct or incorrect antecedent of the Anaphoric *one*, as indicated in column 14. For example, in texts **ID 1 and 4**, **antecedent B** is the **incorrect antecedent** (marked as "**FALSE**" in Column 14) of the Anaphoric *one*, whereas in texts **ID 2, 3, and 5**, **antecedent B** is the **correct antecedent** (marked as "**TRUE**" in Column 14).
- **Columns 11 & 12:** Marks the start and end character offsets of antecedent B.
- **Column 13:** Marks whether antecedent A is the correct antecedent of the Anaphoric *one* or not. For example, in texts **ID 1 and 4**, **antecedent A** is the **correct antecedent** (marked as "**TRUE**"), whereas in texts **ID 2, 3, and 5**, **antecedent A** is **not the correct antecedent** (marked as "**FALSE**") of the Anaphoric *one*.
- **Column 14:** Marks whether antecedent B is the correct antecedent of the Anaphoric *one* or not. For example, in texts **ID 1 and 4**, **antecedent B** is the **incorrect antecedent** (marked as "**FALSE**"), whereas in texts **ID 2, 3, and 5**, **antecedent B** is the **correct antecedent** (marked as "**TRUE**") of the Anaphoric *one*.
- **Column 15:** Marks the linear distance between antecedent A and the Anaphoric *one*. "**Local**" means that **antecedent A** is **closer** to the Anaphoric *one* in linear distance than antecedent B, as seen in text **ID 1**. In contrast, "**non-local**" means that **antecedent A** is **farther** from the Anaphoric *one* in linear distance than antecedent B, as seen in texts **ID 2, 3, 4, and 5**.
- **Column 16:** Marks the linear distance between antecedent B and the Anaphoric *one*. In text **ID 1**, **antecedent B** is **farther** (marked as "**non-local**") from the Anaphoric *one* in linear distance than antecedent A, whereas in texts **ID 2, 3, 4, and 5**, it is **closer** (marked as "**local**") to the Anaphoric *one*.
- **Column 17:** Marks the number feature of the correct antecedent. "**Npsg**" means that the correct antecedent is [**+singular**], as seen in texts **ID 1, 4, and 5**. In contrast, "**nppl**" indicates that the antecedent is [**+plural**], as in texts **ID 2 and 3**.
- **Column 18:** Marks the category of the phrase that introduces the Anaphoric *one* (a.k.a. licenser).
- **Column 19:** Presents the URL of the raw text (Wikipedia website link).

1.2 The D(istance)F(actor)AO Sub-corpus

Table A.2

Examples from the DFAO sub-corpus

column 1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19
ID	text	one/ones	antecedent type	one-start character offset	one-end character offset	A	antecedent A-start character offset	antecedent A-end character offset	B	antecedent B-start character offset	antecedent B-end character offset	A-antecedent	B-antecedent	A-antecedent	B-antecedent	antecedent number	one-licensor category	URL
2	When, at age 90, he was asked about the making of this short and if there had been any further film appearances , he responded (with mock incredulity), "I don't remember making THIS one!"	one	discourse	356	358	film appearances	270	285	short	220	234	FALSE	TRUE	local	non-local	npsg	det	https://en.wikipedia.org/wiki/Cayford_Carter
3	The real significance of that was that when he came back to the pits he told me what happened and said "I'm sure that the wall moved!" And even though I've heard every excuse every driver has ever made, I certainly hadn't heard of that one!	one	linguistic	236	238	excuse	168	173	driver	181	186	TRUE	FALSE	non-local	local	npsg	det	https://en.wikipedia.org/wiki/Ayrton_Senna
4	In 2013 she released "The Funambulist" (Ugly Cat Music), a concept album about life in between two continents and cultures. "The Funambulist" was enthusiastically received by audience and critics. Dave Marsh said about her: "The best and scariest thing about Giulia Millanta is not just that she is truly a tightrope walker herself but that she continually pulls you out there with her... with melodies, singing, scores, imagery. No matter what language her lyrics may be in... and I've lost	one	discourse	655	657	art	598	600	album	67	71	FALSE	TRUE	local	non-local	npsg	adj	https://en.wikipedia.org/wiki/Giulia_Millanta
5	A request to repeat a performance , as in <i>Encore!</i> , lit. 'again'; also used to describe additional songs played at the end of a gig . Francophones would say «Une autre!» ('Another one!') or «Bis!» to request «un rappel» or «un bis».	one	discourse	176	178	gig	126	128	performance	21	31	FALSE	TRUE	local	non-local	npsg	det	https://en.wikipedia.org/wiki/Glossary_of_French_expressions_in_English

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Note. The DFAO sub-corpus shares the same format and labeling as the GAO corpus. The only difference is that **in the DFAO sub-corpus, the incorrect antecedent (either antecedent A or B) is always closer (local) to the Anaphoric *one*** in linear distance than the correct one. In other words, the column labeled "FALSE" always corresponds to the column labeled "local." For instance, in texts ID 2, 4, and 5 within the above examples, **antecedent A** is the **incorrect antecedent** (marked as "FALSE" in Column 13) and is closer (marked as "local" in Column 15) to the Anaphoric *one* in linear distance than antecedent B, which is the correct antecedent of the Anaphoric *one*. In contrast, in text ID 3, **antecedent B** is the **incorrect antecedent** (marked as "FALSE" in Column 14) and is closer (marked as "local" in Column 16) to the Anaphoric *one* in linear distance than antecedent A, which is the correct antecedent of the Anaphoric *one*. For easier reading, we used different colors to present the distinctions. Words in green font represent the correct antecedent, and the corresponding "true" and "non-local" values are also marked in green font. In contrast, words in blue font represent the incorrect antecedent, and the corresponding "false" and "local" values are also marked in green font.

1.3 The N(umber)F(actor)AO Sub-corpus

Table A.3

Examples from the NFAO sub-corpus

column 1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19
ID	text	one/ones	antecedent type	one-start character offset	one-end character offset	A	antecedent A-start character offset	antecedent A-end character offset	B	antecedent B-start character offset	antecedent B-end character offset	A-antecedent	B-antecedent	A-antecedent	B-antecedent	antecedent number	one-licensor category	URL
6	Silvio Manelli presented a magic wand to the Pope John Paul II in 2002 and petitioned the Pope to proclaim St. Don Bosco the Patron of Stage Magicians to which the Pope responded "You'll need a lot of magic wands to change our world ; but let's make a start with this one !"	one	linguistic	373	375	magic wands	307	317	world	10	14	TRUE	FALSE	mismatch	match	nppl	det	https://en.wikipedia.org/wiki/Silvio_Manelli
7	When the player reaches the correct time period, the player must then fight through the end of the level and defeat the level boss to rescue the captured soldier . Unlimited continues are available on all levels except the last one .	one	linguistic	227	229	levels	204	209	soldier	154	160	TRUE	FALSE	mismatch	match	nppl	adj	https://en.wikipedia.org/wiki/Time_Soldiers
8	It is perhaps the easiest to understand TDI devices by contrast with more well-known types of CCD sensors . The best known is the staring array one . In it, there are hundreds or thousands of adjacent rows of specially engineered semi conductor which react to light by accumulating charge.	one	discourse	143	145	sensors	98	104	semi conductor	228	240	TRUE	FALSE	mismatch	match	nppl	adj	https://en.wikipedia.org/wiki/Time_delay_and_integration
9	The Labour Party obtained a result that brought it back into the position of "the largest Norwegian party , not just the largest of many middle-sized ones". The Progress Party obtained its best result ever, surpassing the Conservatives by a large margin and becoming almost as large as the parties in the Bondevik government combined.	ones	linguistic	149	152	Norwegian party	90	104	position	65	72	TRUE	FALSE	mismatch	match	nppl	adj	https://en.wikipedia.org/wiki/2005_Norwegian_parliamentary_election
10	In the 4th quarter, with it only being a 6-point game, Mark Dantonio called a fake punt on 4th and 4 as punter Mike Sadler ran for 25 yards to the Iowa 38 yard line to pick up the first down. As with most Spartan trick plays , coach Dantonio coined this one . "Hey diddle, diddle, send Sadler up the middle".	one	linguistic	253	255	Spartan trick plays	205	223	coach	226	230	TRUE	FALSE	mismatch	match	nppl	det	https://en.wikipedia.org/wiki/2013_Michigan_State_Spartans_football_season

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Note. The NFAO sub-corpus also shares an identical format and labeling with the GAO corpus. The only difference is that in the NFAO sub-corpus, the columns marking whether the antecedent is local or nonlocal to the Anaphoric *one* are replaced with columns marking whether the antecedent agrees or disagrees with the Anaphoric *one* in number. **In the NFAO sub-corpus, the incorrect antecedent (either antecedent A or B) always matches the Anaphoric *one* in number, while the correct one does not.** In other words, the column labeled "FALSE" always corresponds to the column labeled "match." For instance, in the above examples, **antecedent B** is the **incorrect antecedent** (marked as "**FALSE**" in Column 14) and agrees with the Anaphoric *one* in number (marked as "**match**" in Column 16), whereas the correct antecedent (antecedent A, which is marked as "**TRUE**" in Column 13) does not (marked as "**mismatch**" in Column 15). For easier reading, we used different colors to show the distinctions. Words in green font represent the correct antecedent, and the corresponding "true" and "mismatch" values are also marked in green font. In contrast, words in blue font represent the incorrect antecedent, and the corresponding "false" and "match" values are also marked in blue font.

2. The NPE Corpora

2.1 The G(eneral)NPE Corpus

Table A.4

Examples from the GNPE corpus

column 1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
ID	text	licensor	licensor-start character offset	licensor-end character offset	A	antecedent A-start character offset	antecedent A-end character offset	B	antecedent B-start character offset	antecedent B-end character offset	A-antecedent	B-antecedent	A-antecedent	B-antecedent	antecedent number	licensor-category	URL
1	The first three of these locomotives were later described as the Clara Class, while the fourth was included in this Class by some and included in the subsequent Scotia Class by others.	fourth	88	93	locomotives	25	35	Clara Class	65	75	TRUE	FALSE	non-local	local	nppl	det	https://en.wikipedia.org/wiki/D-6-2
2	That's why I want to find a different path, instead of fighting over our beliefs... There's not much time, but there is some! And I can't give up now...!	some	120	123	beliefs	73	79	time	101	104	FALSE	TRUE	non-local	local	npsg	det	https://en.wikipedia.org/wiki/ShinMegami-Iensei:_Devil_Survivor_2
3	Marty Links' Emmy Lou (November 14, 1966): "I suppose it's called 'mad money' because fathers always get so mad when you ask for some !"	some	129	132	fathers	86	92	money	71	75	FALSE	TRUE	local	non-local	npsg	det	https://en.wikipedia.org/wiki/Marty_Links
4	Moe then pours honey and ketchup on bread, declaring "If there's anything I like better than honey and ketchup, it's bologna and whipped cream--and we haven't got any !"	any	162	164	bologna	117	123	ketchup	103	109	TRUE	FALSE	local	non-local	npsg	det	https://en.wikipedia.org/wiki/Idiot's_Deluxe
5	Moe then pours honey and ketchup on bread, declaring "If there's anything I like better than honey and ketchup, it's bologna and whipped cream--and we haven't got any !"	any	162	164	whipped cream	129	141	ketchup	103	109	TRUE	FALSE	local	non-local	npsg	det	https://en.wikipedia.org/wiki/Idiot's_Deluxe

Note. The annotation method in the NPE corpora is nearly identical to that in the AO corpora. Since NPE can only have linguistic antecedents (Hankamer and Sag 1976; Sag and Hankamer 1984), the column labeling "antecedent type" (Column 4) in the AO corpora is omitted in the NPE corpora. And the two NPE corpora share the same annotation format, in which a total of 18 columns are included:

- **Column 1:** Marks the text number. "1" means that the text is the first example in the GNPE corpus. The above example lists the first 5 texts in the GNPE corpus.
- **Column 2:** Presents the raw text collected from English Wikipedia. The word in bold represents the NPE licenser, and the underlined word represents the correct antecedent of the NPE.
- **Column 3:** Marks the NPE licenser. As the site for NPE is empty, we mark the licenser so that the models can detect the NPE site.
- **Columns 4 & 5:** Marks the start and end character offsets of the NPE licenser.
- **Column 6:** Marks the first antecedent candidate. It can be either the correct or incorrect antecedent of the NPE, as indicated in Column 12. For example, in texts **ID 1, 4, and 5**, **antecedent A** is the **correct antecedent** (marked as "**TRUE**" in Column 12) of the NPE, whereas in texts **ID 2 and 3**, **antecedent A** is **not the correct antecedent** (marked as "**FALSE**" in Column 12).
- **Columns 7 & 8:** Marks the start and end character offsets of antecedent A.
- **Column 9:** Marks the second antecedent candidate. It can either be the correct or incorrect antecedent of the NPE, as indicated in column 13. For example, in texts **ID 1, 4, and 5**, **antecedent B** is the **incorrect antecedent** (marked as "**FALSE**" in Column 13) of the NPE, whereas in texts **ID 2 and 3**, **antecedent B** is the **correct antecedent** (marked as "**TRUE**" in Column 13).
- **Columns 10 & 11:** Marks the start and end character offsets of antecedent B.
- **Column 12:** Marks whether antecedent A is the correct antecedent of the NPE or not. For example, in texts **ID 1, 4, and 5**, **antecedent A** is the **correct antecedent** (marked as "**TRUE**"), whereas in texts **ID 2 and 3**, **antecedent A** is **not the correct antecedent** (marked as "**FALSE**") of the NPE.
- **Column 13:** Marks whether antecedent B is the correct antecedent of the NPE or not. For example, in texts **ID 1, 4, and 5**, **antecedent B** is the **incorrect antecedent** (marked as "**FALSE**"), whereas in texts **ID 2 and 3**, **antecedent B** is the **correct antecedent** (marked as "**TRUE**") of the NPE.
- **Column 14:** Marks the linear distance between antecedent A and the NPE licenser. "**Local**" means that **antecedent A** is **closer** to the NPE licenser in linear distance than antecedent B, as seen in texts **ID 3, 4, and 5**. In contrast, "**non-local**" means that **antecedent A** is **farther** from the NPE licenser in linear distance than antecedent B, as seen in texts **ID 1 and 2**.
- **Column 15:** Marks the linear distance between antecedent B and the NPE licenser. In texts **ID 1 and 2**, **antecedent B** is **closer** (marked as "**local**") to the NPE licenser in linear distance than antecedent A, whereas in texts **ID 3, 4, and 5**, it is **farther** (marked as "**non-local**") from the NPE licenser.
- **Column 16:** Marks the number feature of the correct antecedent. "**Npsg**" means that the correct antecedent is [**+singular**], as seen in texts **ID 2, 3, 4, and 5**. In contrast, "**nppl**" indicates that the antecedent is [**+plural**], as seen in text **ID 1**.
- **Column 17:** Marks the category of the NPE licenser.
- **Column 18:** Presents the URL of the raw text (Wikipedia website link).

2.2 The D(istance)F(actor)NPE Sub-corpus

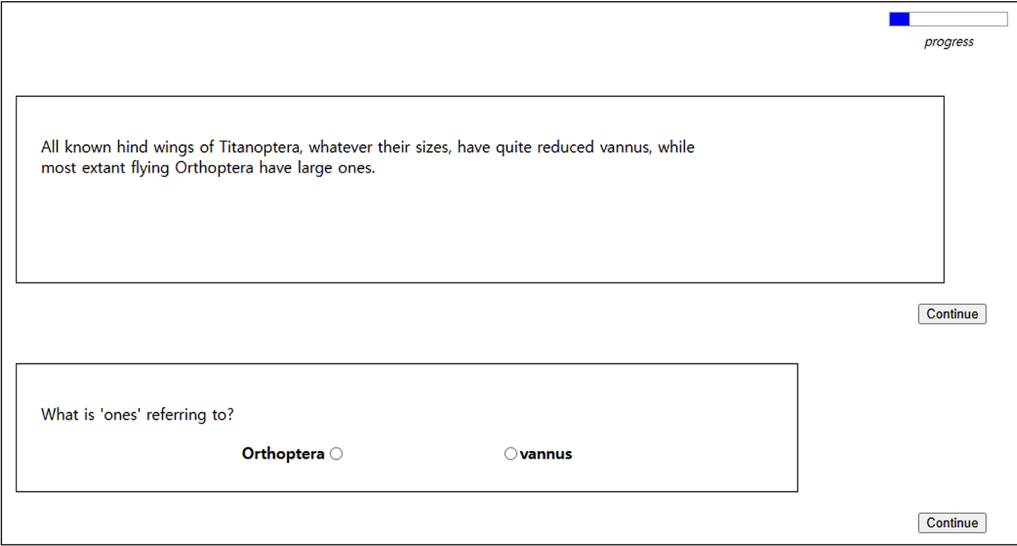
Table A.5

Examples from the GNPE corpus

column 1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
ID	text	licensor	licensor-start character offset	licensor-end character offset	A	antecedent A-start character offset	antecedent A-end character offset	B	antecedent B-start character offset	antecedent B-end character offset	A-antecedent	B-antecedent	A-antecedent	B-antecedent	antecedent number	licensor-category	URL
1	The first three of these <i>locomotives</i> were later described as the <i>Clara Class</i> , while the <i>fourth</i> was included in this Class by some and included in the subsequent Scotia Class by others.	fourth	88	93	<i>locomotives</i>	25	35	<i>Clara Class</i>	65	75	TRUE	FALSE	<i>non-local</i>	<i>local</i>	nppl	det	https://en.wikipedia.org/wiki/0-6-2
2	Marry Links' Emmy Lou (November 14, 1966): "I suppose it's called 'mad <i>money</i> ' because <i>fathers</i> always get so mad when you ask for <i>some</i> !"	some	129	132	<i>fathers</i>	86	92	<i>money</i>	71	75	FALSE	TRUE	<i>local</i>	<i>non-local</i>	npsg	det	https://en.wikipedia.org/wiki/Marry_Links
3	The brothers turned to Jack Kirby. Steve Schanes recalled, "I figured if you want to get people's attention with a new comic book, who better to do it with than the King of Comics, Jack Kirby! We were already friends with Jack. We used to send him <i>free copies of comics</i> he'd drawn for other <i>publishers</i> because they never sent him <i>any</i> !"	any	330	332	<i>publishers</i>	291	300	<i>free copies of comics</i>	248	268	FALSE	TRUE	<i>local</i>	<i>non-local</i>	nppl	det	https://en.wikipedia.org/wiki/Antony_Coelho
4	Fagan began borrowing large amounts from Holiday to support the restaurant. Holiday obliged but soon fell on hard times herself. "I needed some <i>money</i> , one <i>night</i> and I knew Mom was sure to have <i>some</i> ", she said.	some	192	195	<i>night</i>	154	158	<i>money</i>	144	148	FALSE	TRUE	<i>local</i>	<i>non-local</i>	npsg	det	https://en.wikipedia.org/wiki/Billie_Holiday
5	A voice she described as loud and deep, probably not Ogietree's, asked who it was. She responded that she had brought <i>fresh towels</i> , to which the <i>voice</i> said "We don't need <i>any</i> ".	any	171	173	<i>fresh towels</i>	118	129	<i>voice</i>	145	149	TRUE	FALSE	<i>non-local</i>	<i>local</i>	nppl	det	https://en.wikipedia.org/wiki/Murder_of_Artemus_Ogietree

Note. The DFNPE sub-corpus shares the same format and labeling as the GNPE corpus. The only difference is that **in the DFNPE sub-corpus, the incorrect antecedent (either antecedent A or B) is always closer (local) to the NPE licenser** in linear distance than the correct one. In other words, the column labeled "FALSE" always corresponds with the column labeled "local." For instance, in texts ID 1 and 5 within the above examples, antecedent B is the incorrect antecedent (marked as "FALSE" in Column 13) and is closer (marked as "local" in Column 15) to the NPE licenser in linear distance than antecedent A, which is the correct antecedent of the NPE licenser. In contrast, in texts ID 2, 3, and 4, antecedent A is the incorrect antecedent (marked as "FALSE" in Column 12) and is closer (marked as "local" in Column 14) to the NPE licenser in linear distance than antecedent B, which is the correct antecedent of the NPE licenser. For easier reading, we used different colors to present the distinctions. Words in green font represent the correct antecedent, and the corresponding "true" and "non-local" values are also marked in green font. In contrast, words in blue font represent the incorrect antecedent, and the corresponding "false" and "local" values are also marked in green font.

Appendix B Example of an Anaphoric *one* Experiment Presented to Human Participants



progress

All known hind wings of Titanoptera, whatever their sizes, have quite reduced vannus, while most extant flying Orthoptera have large ones.

Continue

What is 'ones' referring to?

Orthoptera ☐ ☐ vannus

Continue

Figure B.1

An example of the Anaphoric *one* experiment presented to human participants.

Appendix C Results from the Forced-choice Experiments for Anaphoric *one* and NPE Resolution Tasks & Experiment with Fine-tuned LLMs and Results

1. Results from the Forced-choice Experiments for Anaphoric *one* and NPE Resolution Tasks

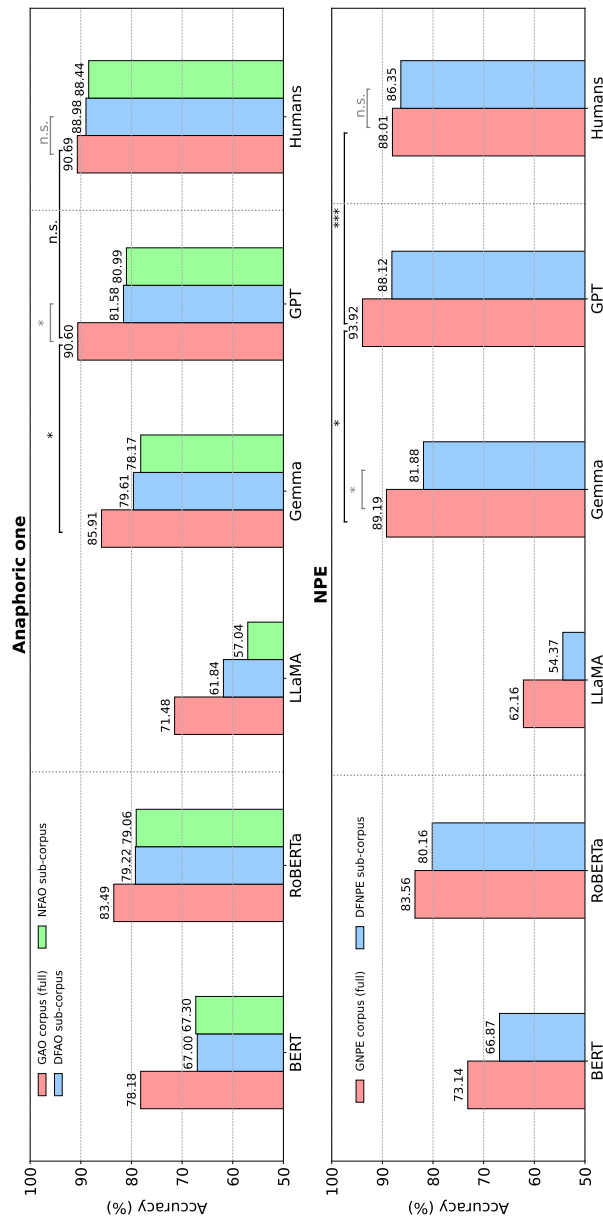


Figure C.1

Results from the forced-choice experiments for Anaphoric *one* and NPE resolution tasks (compared with sub-corpora) using different language models and testing on humans. Statistical significance is indicated only for selected comparisons between the conditions of interest for visual clarity. Asterisks indicate statistically significant differences ($p < 0.05$), while “n.s.” denotes non-significant differences ($p \geq 0.05$).

2. Experiments with Fine-tuned LLMs

The GPT-4o mini model (OpenAI 2024) was fine-tuned using Supervised Fine-Tuning (SFT) via OpenAI's official fine-tuning API (<https://platform.openai.com/docs/guides/fine-tuning>). For this purpose, we used our own AO-NPE datasets, which contain annotated Anaphoric *one* and NPE instances. Once training was completed, a unique fine-tuned model ID was assigned, which could then be used for inference through the Chat Completion API.

Besides the zero-shot setting, we also tested the LLaMA (Meta AI 2024) and Gemma (Gemma Team 2024) models in a fine-tuning setting using the LoRA (Low-Rank Adaptation) technique (Hu et al. 2021), enabling memory-efficient fine-tuning. LoRA works by learning low-rank matrices for a subset of weights, allowing the model to effectively adapt to a specific task with a small number of trainable parameters while preserving the prior knowledge of the pre-trained model. Training was conducted over a total of 10 epochs, and the checkpoint with the best performance was saved and used for evaluation. After training, the inference results of each model were evaluated by comparing the predictions against the ground truth to calculate accuracy. Training was conducted on a system equipped with an NVIDIA RTX 6000 Ada Generation GPU, using PyTorch 2.3.1 and CUDA 11.8. The implementation utilized Huggingface Transformers (Wolf et al. 2019) and PEFT libraries (Mangrulkar and Paul 2023). To ensure a fair comparison of performance across the models, experiments were carried out under conditions that were as similar as possible.⁵ Figure C.2 illustrates the fine-tuning workflow using the LoRA-based method applied in our experiments.

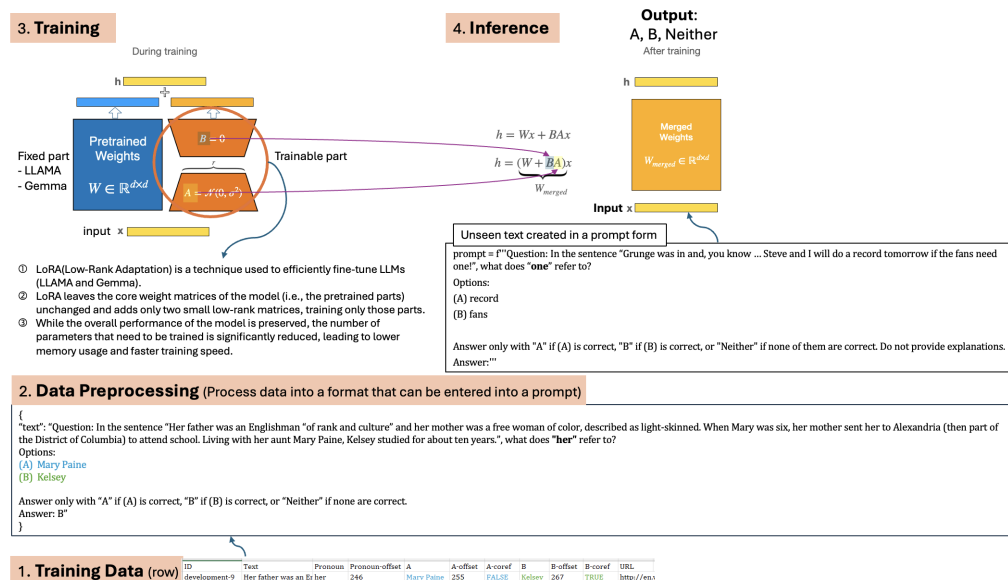


Figure C.2

LoRA-based fine-tuning workflow for the forced-choice anaphora resolution tasks.

3. Experimental Results with Fine-tuned LLMs

We also evaluated LLMs in this fine-tuned setting. The results show that training strategy influenced model performance. In particular, fine-tuned LLaMA showed improvement compared to its zero-shot results, whereas Gemma and GPT showed little or no gain from fine-tuning. These findings suggest that model performance depended on fine-tuning strategies and model architectures.

⁵ All code, annotated datasets, and experiment scripts are available on Github (<https://github.com/nayoun1/Anaphoric-One-and-NPE>)

Table C.1

Accuracy of antecedent resolution from BERT, RoBERTa, fine-tuned LLMs, and humans.

Test data	BERT	RoBERTa	LLaMA	Gemma	GPT	Human
GAO corpus	78.18%	83.49%	74.0%	77.67%	83.33%	90.69%
DFAO	67.00%	79.22%	81.17%	71.43%	76.62%	88.98%
NFAO	67.30%	79.06%	67.83%	61.54%	67.13%	88.44%
GNPE corpus	73.14%	83.56%	84.0%	78.0%	83.67%	88.01%
DFNPE	66.87%	80.16%	76.83%	73.78%	75.61%	86.35%

Appendix D Examples of Manual Re-examination of LLM Responses and Statistical Analysis of LLMs vs. Humans in the Open-ended Anaphoric *one* and NPE Resolution Experiment

1. Manual Reexamination on the Responses of Three LLMs in the Open-ended Anaphoric *one* Resolution Task

Table D.1Examples of manual re-examination on LLM responses of the predicted antecedent of Anaphoric *one*. Answers reclassified as “TRUE” after re-examination are marked in pink highlights.

text_id	text_corpus	A	B	A-antecedent	B-antecedent	human_a	correct_hum	expected_ans	gpt_answer	correct_C	correct_C	llama_answer	correct_LLa	recheck	Gemma_LoR	correct_Gem	recheck	dataset
1	story, may your careers be	war	police	TRUE	FALSE	1	TRUE	war	WAR	TRUE	TRUE	The citizens	FALSE	FALSE	you	FALSE	FALSE	gao
2	that these officers, did not die in	officers	people	FALSE	TRUE	0.666666	TRUE	people	PEOPLE	TRUE	TRUE	The people	FALSE	TRUE	the people	FALSE	TRUE	gao
3	friend of mine who wasn't a	songwriter	lines	FALSE	TRUE	1	TRUE	lines	LINES	TRUE	TRUE	lines	TRUE	TRUE	lines	TRUE	TRUE	gao
4	what, that's OK, 'cause music needs	record	fans	TRUE	FALSE	1	TRUE	record	RECORD	TRUE	TRUE	fansExplanation	FALSE	FALSE	record	TRUE	TRUE	gao
5	in American Record Guide was	review	reading	FALSE	TRUE	0.5	TRUE	reading	READING	TRUE	TRUE	Cooke's playing	FALSE	FALSE	reading	TRUE	TRUE	gao
6	sequence with singer Conje	film appear	short	FALSE	TRUE	0.333333	FALSE	short	SHORT	TRUE	TRUE	The short	FALSE	TRUE	short	TRUE	TRUE	gao
7	satisfied with his ability to hit the	enemy	target	TRUE	FALSE	1	TRUE	enemy	ENEMY	TRUE	TRUE	Elman	FALSE	FALSE	target	FALSE	FALSE	gao
8	that when he came back to the pits	driver	excuse	TRUE	FALSE	1	TRUE	excuse	EXCUSE	TRUE	TRUE	The wall	FALSE	FALSE	that one	FALSE	FALSE	gao
9	and squirrel" (as he refers to them)	plan	scheme	TRUE	FALSE	1	TRUE	plan	PLAN	TRUE	TRUE	Boris	FALSE	FALSE	plan	TRUE	TRUE	gao
10	Funambulist" (Judy Cat Music), a	art	album	FALSE	TRUE	1	TRUE	album	ALBUM	TRUE	TRUE	Giulia Millanta's	FALSE	FALSE	her next one	FALSE	FALSE	gao
11	as in Encore!, lit. 'again' also used	sig	performa	FALSE	TRUE	0.666666	TRUE	performance	SONG	FALSE	FALSE	One (the perfor	FALSE	FALSE	performance	TRUE	TRUE	gao
12	humorously in exclamations when	exclamations	shot	FALSE	TRUE	1	TRUE	shot	SHOT	TRUE	TRUE	The shot	FALSE	TRUE	shot	TRUE	TRUE	gao
13	music band Marillion's music	statement	biscuits	FALSE	TRUE	1	TRUE	biscuits	BISCUIT	FALSE	TRUE	Jacko	FALSE	FALSE	biscuit	FALSE	TRUE	gao
14	owning multiple cats for	car	househol	TRUE	FALSE	1	TRUE	car	CAR	TRUE	TRUE	one driver	FALSE	FALSE	one driver	FALSE	FALSE	gao
15	23,000 copies, making it one of	game	enjoyment	TRUE	FALSE	1	TRUE	game	GORGON	FALSE	FALSE	Nasir Gebelli	FALSE	FALSE	Gorgon	FALSE	FALSE	gao
16	falling apart, so it's time for a new	blanket	time	TRUE	FALSE	1	TRUE	blanket	BLANKET	TRUE	TRUE	a new blanket	FALSE	FALSE	blanket	TRUE	TRUE	gao
17	Clifford by making him a new	blanket	Clifford	FALSE	TRUE	1	TRUE	blanket	BLANKET	TRUE	TRUE	Clifford	FALSE	FALSE	blanket	TRUE	TRUE	gao
18	the conservative memoirist, the	fire	laws	TRUE	TRUE	1	TRUE	laws	LAWS	TRUE	TRUE	Greek nation	FALSE	FALSE	laws	TRUE	TRUE	gao
19	demobilizes and comes back to his	sailor	shepherd	TRUE	FALSE	1	TRUE	shepherd	GIRL	FALSE	FALSE	girl	FALSE	FALSE	girl	FALSE	FALSE	gao
20	sent Lord Elgin a sentimental	acknowledge	Queen	TRUE	FALSE	1	TRUE	acknowledge	ACKNOWLEDGE	TRUE	TRUE	The Queen	FALSE	FALSE	acknowledge	TRUE	TRUE	gao
21	had decided on the importance of	names	places	TRUE	FALSE	1	TRUE	names	NAMES	TRUE	TRUE	these places	FALSE	FALSE	names	TRUE	TRUE	gao
22	awarded the Iron Cross First Class	victory	confirmat	TRUE	FALSE	1	TRUE	victory	VICTORY	TRUE	TRUE	Our own group	FALSE	FALSE	victory	TRUE	TRUE	gao
23	the 'great and the good' who	Director	codes	TRUE	FALSE	1	TRUE	Director	DIRECTOR	TRUE	TRUE	Director/The pre	FALSE	TRUE	a GAY one	FALSE	FALSE	gao
24	vulcanization oven where they	nails	nose-piec	FALSE	TRUE	1	TRUE	nose-pieces	MASK	FALSE	FALSE	One	FALSE	FALSE	mask	FALSE	FALSE	gao
25	began an affair with Charles	influence	baby	FALSE	TRUE	1	TRUE	baby	BABY	TRUE	TRUE	a baby	FALSE	TRUE	baby	TRUE	TRUE	gao
26	second single from Get Happy!!	routine	single	FALSE	TRUE	0.666666	TRUE	single	SINGLE	TRUE	TRUE	The single	FALSE	TRUE	music video	FALSE	FALSE	gao
27	windowing system allowing	game collect	Amiga ve	FALSE	TRUE	1	TRUE	Amiga versio	GAME	FALSE	FALSE	Info magazine	FALSE	FALSE	the game	FALSE	FALSE	gao
28	across the country in East Lansing,	boys	trimming	FALSE	TRUE	1	TRUE	trimming	THING	FALSE	FALSE	The Corvallis b	FALSE	FALSE	One thing	FALSE	FALSE	gao

2. Manual Reexamination on the Responses of Three LLMs in the Open-ended NPE Resolution Task

Table D.2

Examples of manual re-examination of LLM responses of the predicted antecedent of NPE. Answers reclassified as “TRUE” after re-examination are marked in pink highlights.

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text_id	text_corpus	license	A	B	A-antecedent	B-antecedent	target	expected	ans	gpt_answer	correct_GPT	recheck	llama_answer	correct_LL	llama_recheck	Gemma_LoR	correct_Gem	Gem_recheck	dataset
1	The first three	fourth	locomotives	Clara Class	TRUE	FALSE	fourth	locomotives	THE FOURTH	FALSE	FALSE	FALSE	The fourth	FALSE	FALSE	locomotives	TRUE	TRUE	gnpe
2	That's why I	some	beliefs	time	FALSE	TRUE	some	time	TIME	TRUE	TRUE	TRUE	Answer: I	FALSE	FALSE	time	TRUE	TRUE	gnpe
3	Marty Link's	some	fathers	money	FALSE	TRUE	some	money	MAD MONEY	FALSE	FALSE	FALSE	Marty Link's	FALSE	FALSE	money	TRUE	TRUE	gnpe
4	Moe then put	any	bologna	ketchup	TRUE	FALSE	any	bologna	BOLOGNA /	FALSE	TRUE	TRUE	Moe	FALSE	FALSE	bologna and	FALSE	TRUE	gnpe
5	Moe then put	any	whipped crea	ketchup	TRUE	FALSE	any	whipped crea	BOLOGNA /	FALSE	TRUE	TRUE	Moe	FALSE	FALSE	bologna and	FALSE	TRUE	gnpe
6	The brothers	any	publishers	free copies of	FALSE	TRUE	any	free copies of	COMICS	FALSE	FALSE	FALSE	Jack Kirby	FALSE	FALSE	copies of con	FALSE	TRUE	gnpe
7	Fagan began	some	night	money	FALSE	TRUE	some	money	MONEY	TRUE	TRUE	TRUE	Holiday	FALSE	FALSE	money	TRUE	TRUE	gnpe
8	A voice she d	any	fresh towels	voice	TRUE	FALSE	any	fresh towels	FRESH TOW	TRUE	TRUE	TRUE	She	FALSE	FALSE	towels	FALSE	TRUE	gnpe
9	He refused be	any	honor	artist	TRUE	FALSE	any	honor	HONOR	TRUE	TRUE	TRUE	Ferr / ©	FALSE	FALSE	honor	TRUE	TRUE	gnpe
10	A subsequent	any	manufacturin	cartoonish en	TRUE	FALSE	any	manufacturin	MANUFACT	TRUE	TRUE	TRUE	Australia's fo	FALSE	FALSE	manufacturin	TRUE	TRUE	gnpe
11	The announce	any	company	ships	FALSE	TRUE	any	ships	SHIPS	TRUE	TRUE	TRUE	Seaborn Fre	FALSE	FALSE	ships	TRUE	TRUE	gnpe
12	Privacy mind	any	engines	data	FALSE	TRUE	any	data	DATA	TRUE	TRUE	TRUE	Google or Ya	FALSE	FALSE	data	TRUE	TRUE	gnpe
13	Featuring Joh	any	printing	title	FALSE	TRUE	any	title	OFFICIAL T	FALSE	FALSE	FALSE	John and Gill	FALSE	FALSE	title	TRUE	TRUE	gnpe
14	An advantag	any	file	data	FALSE	TRUE	any	data	DATA	TRUE	TRUE	TRUE	the file	FALSE	FALSE	data	TRUE	TRUE	gnpe
15	On Day 64, h	any	money	group	TRUE	FALSE	any	money	FL / AVIAS	FALSE	FALSE	FALSE	Daniel & Lib	FALSE	FALSE	money	TRUE	TRUE	gnpe
16	Whether then	any	meaning	sentence	TRUE	FALSE	any	meaning	MEANING	TRUE	TRUE	TRUE	B	FALSE	FALSE	meaning	TRUE	TRUE	gnpe
17	He appears in	any	cryptic advic	Plumber	TRUE	FALSE	any	cryptic advic	CRYPTIC AI	TRUE	TRUE	TRUE	Nevo	FALSE	FALSE	advice	FALSE	TRUE	gnpe
18	Part-time job	his	family incom	earning powe	FALSE	TRUE	his	earning powe	THE MAN	FALSE	FALSE	FALSE	man	FALSE	FALSE	man	FALSE	TRUE	gnpe
19	Some found i	his	Oppressor	reward	FALSE	TRUE	his	reward	OPPRESSOR	FALSE	FALSE	FALSE	Oppressor	FALSE	FALSE	Oppressor	FALSE	TRUE	gnpe
20	He was empl	his	ships	master	TRUE	FALSE	his	ships	THE SUCC	FALSE	FALSE	FALSE	Master, Mast	FALSE	FALSE	ship	FALSE	TRUE	gnpe
21	Her ability t	his	exploration o	good acts	TRUE	FALSE	his	exploration o	HAWTHORN	FALSE	FALSE	FALSE	Hawthorne	FALSE	FALSE	Hawthorne	FALSE	TRUE	gnpe
22	One was pres	his	guitar	note	TRUE	FALSE	his	guitar	ERIC CLAP	FALSE	FALSE	FALSE	Clapton	FALSE	FALSE	Clapton	FALSE	TRUE	gnpe
23	Soon after, M	Tom's	child	months	TRUE	FALSE	Tom's	child	TOM	FALSE	FALSE	FALSE	Tom's	FALSE	FALSE	child	TRUE	TRUE	gnpe
24	In 1788 Rob	some	boiled egg	way	TRUE	FALSE	some	boiled egg	BOILED EGG	TRUE	TRUE	TRUE	Burns	FALSE	FALSE	boiled egg	TRUE	TRUE	gnpe
25	The greetin	some	figgy puddin	audience	TRUE	FALSE	some	figgy puddin	FIGGY PUDD	TRUE	TRUE	TRUE	figgy puddin	TRUE	TRUE	figgy puddin	TRUE	TRUE	gnpe
26	Kesha is 'sen	some	success	name	TRUE	FALSE	some	success	HER SUCC	FALSE	FALSE	FALSE	Taylor Swift	FALSE	FALSE	success	TRUE	TRUE	gnpe
27	Saganth also	some	scenes	novice	TRUE	FALSE	some	scenes	SCENES	TRUE	TRUE	TRUE	Chowdhury,	FALSE	FALSE	scenes	TRUE	TRUE	gnpe
28	Baldwin resp	some	childhood	things	TRUE	FALSE	some	childhood	TIME	FALSE	FALSE	FALSE	Baldwin, her	FALSE	FALSE	childhood	TRUE	TRUE	gnpe
29	When INAH	some	archaeologica	maintenance	TRUE	FALSE	some	archaeologica	ARCHAEOLOG	TRUE	TRUE	TRUE	The petrogly	FALSE	FALSE	archaeologica	TRUE	TRUE	gnpe

Table D.3

Statistics for comparisons among humans and models for the open-end antecedent identification task. Statistical methods and notations are the same as Table 10.

Dataset	Human (W)			GPT (χ^2)		Gemma (χ^2)
	GPT	Gemma	LLaMA	Gemma	LLaMA	LLaMA
GAO corpus	1,094.0 ($< .001$)	781.0 ($< .001$)	292.0 ($< .001$)	34.46 ($< .001$)	104.36 ($< .001$)	49.42 ($< .001$)
DFAO	521.0 ($< .001$)	344.5 ($< .001$)	117.0 ($< .001$)	7.84 (0.005)	40.88 ($< .001$)	23.59 ($< .001$)
NFAO	178.0 ($< .001$)	178.0 ($< .001$)	9.0 ($< .001$)	9.03 (0.003)	48.43 ($< .001$)	22.69 ($< .001$)
GNPE corpus	2,211.0 ($< .001$)	2,076.5 ($< .001$)	280.5 ($< .001$)	0.11 (0.73)	168.27 ($< .001$)	157.21 ($< .001$)
DNPE	658.5 ($< .001$)	651.5 ($< .001$)	273.5 ($< .001$)	0.73 (0.39)	65.33 ($< .001$)	53.64 ($< .001$)

Table D.4

Statistics for comparisons between datasets with different linguistic factors (distance and number) for each model. Statistical methods and notations are the same as Table 11.

LLMs/Human	GAO		GNPE
	DFAO	NFAO	DFNPE
LLaMA	0.11(0.91)	1.22(0.22)	-1.04(0.30)
Gemma	0.32(0.75)	0.84(0.40)	3.17(0.0015)
GPT	1.43(0.15)	1.75(0.08)	2.59(0.0099)
Human	1.18(0.24)	-0.37(0.71)	1.93(0.053)

Appendix E Comparisons between Subcategorized Items: Statistics and Other Strategies

1. Statistics for Comparisons between Parallelism Resolution Strategies

We fit post-hoc generalized linear mixed-effects (GLMM) models to examine whether LLMs and humans' performance differed depending on whether items could be resolved using parallelism strategies (in the *lme4* style: $\text{isCorrect} \sim \text{model} * \text{parallelism} + (1 \mid \text{itemID})$). Human performance was binarized using 0.5 as the cutoff value. In Table E.1 we report estimates (in log-odds), credible intervals, z -values and p -values estimated by the *brms* and *lme4* packages in R (Bates et al. 2015; Bürkner 2017), with a focus on marginal effects of parallelism for each model, estimated by *emmeans* with p -values adjusted using the Tukey method (Lenth and Piskowski 2025). Full detailed annotations on parallelism can be found on GitHub (<https://github.com/nayoun1/Anaphoric-One-and-NPE>).

Table E.1

Exploratory tests of parallelism strategy on model performance in the forced-choice task.

Dataset	Models	Estimates	SE	95% CrI	z	p
GAO	LLaMA	-0.50	0.32	[-1.17, 0.11]	-1.58	0.12
	Gemma	-0.80	0.40	[-1.69, -0.07]	-2.01	0.04
	GPT	-1.38	0.52	[-2.53, -0.42]	-2.68	0.007
	Human	-0.73	0.65	[-2.15, 0.53]	-1.13	0.26
GNPE	LLaMA	0.30	0.42	[-0.52, 1.24]	0.72	0.47
	Gemma	-1.96	0.50	[-3.14, -1.05]	-3.97	< 0.001
	GPT	-1.63	0.59	[-2.95, -0.51]	-2.78	0.006
	Human	-1.20	0.67	[-2.66, 0.11]	-1.79	0.07

2. Annotation and Comparisons between Plausibility Strategies

Plausibility was assessed using ratings of minimal clauses containing the target anaphoric element, with each antecedent candidate substituted into the anaphoric position (see Table E.2 for an example). Using the total of 300 instances from both the GAO and GNPE corpora on which the models were evaluated, an initial screening was conducted by two of the authors to exclude sentences in which only the true antecedent was semantically plausible. The remaining sentences were included in the rating task, yielding 162 items (54%) for *one* and 125 items (42%) for NPE. Each clause, either instantiated with the true antecedent or with the distractor, was independently rated by 5 naive participants (native speakers of English recruited via Prolific) on a 7-point Likert scale (1 = totally implausible, 7 = totally plausible). The results showed that the true antecedent was rated as more plausible than the distractor for *one* (true antecedent: $M = 5.10$, $SD = 2.01$; distractor: $M = 4.60$, $SD = 2.15$; $t = 4.26$, $p < .001$) and NPE (true antecedent: $M = 5.19$, $SD = 1.90$; distractor: $M = 4.67$, $SD = 1.92$; $t = 3.43$, $p < .001$). Based on the ratings, each item was labeled as "Target preferred" if it failed the initial screening or if the mean plausibility rating was higher for the target than for the distractor; otherwise, it was labeled as "Distractor preferred." As a result, 84% of items for *one* and 85% of the items for NPE were "Target preferred."

Unlike parallelism-based strategies, humans consistently took advantage of the plausibility of the antecedent in resolving both Anaphoric *one* and NPE, as evidenced by reduced error rates if the true antecedent rather than the distractor was preferred given the local context, whereas LLMs' behaviors varied (Table E.3). GPT did not appear to benefit substantially from plausibility, as suggested by increased error rates for *one* and similar error rates for NPE. Likewise, Gemma also exhibited higher error rates if the true antecedent was more plausible, a pattern opposite to that observed in humans, while for NPE the pattern was reversed and became similar to humans. By contrast, LLaMA patterned with human participants, showing higher accuracy when the true

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The Mechanism behind Understanding Anaphoric *one* and NPE**Table E.2**

Example of Plausibility Assessment Experiment for Antecedent Options

Item	Example
Raw text	(Text omitted) I'm sure that the wall moved! And even though I've heard every excuse every driver has ever made, I certainly hadn't heard of that <i>one</i> ! (Text omitted) (https://en.wikipedia.org/wiki/Ayrton_Senna)
Target Minimal Clause	I certainly hadn't heard of that <i>one</i> .
Antecedent options	excuse, driver
Plausibility test sentences	I certainly hadn't heard of that excuse . I certainly hadn't heard of that driver .

antecedent, rather than a non-antecedent distractor, was more plausible. None of these effects reached statistical significance, though (see Table E.4).

Table E.3

Error rates of antecedent resolution from zero-shot LLMs and humans, split by whether the true antecedent or the distractor is preferred.

Dataset	Plausibility	LLaMA	Gemma	GPT	Human
GAO	Distractor preferred (16%)	29.79%	8.51%	6.38%	11.35%
	Target preferred (84%)	28.17%	15.48%	8.90%	8.27%
	Difference	1.61%	-6.97%	-3.54%	2.45%
GNPE	Distractor preferred (15%)	47.73%	18.18%	6.82%	19.51%
	Target preferred (85%)	36.11%	9.52%	5.95%	10.68%
	Difference	11.62%	8.66%	0.87%	8.83%

Table E.4

Exploratory tests of antecedent plausibility on model performance in the forced-choice task.

Dataset	Models	Estimates	SE	95% CrI	<i>z</i>	<i>p</i>
GAO	LLaMA	-0.10	0.44	[-0.99, 0.82]	-0.24	0.81
	Gemma	0.76	0.62	[-0.37, 2.23]	1.22	0.22
	GPT	0.51	0.70	[-0.71, 2.24]	0.74	0.46
	Human	-1.02	0.69	[-2.42, 0.43]	-1.47	0.14
GNPE	LLaMA	-0.67	0.44	[-1.54, 0.25]	-1.50	0.13
	Gemma	-0.89	0.54	[-2.02, 0.23]	-1.65	0.10
	GPT	-0.17	0.74	[-1.64, 1.50]	-0.23	0.82
	Human	-0.64	0.76	[-2.11, 1.10]	-0.84	0.40

Appendix F Examples and Results from the Simplified AO-NPE Datasets

1. The Simplified AO-NPE datasets

The simplified AO datasets were created to reduce the structural and distributional complexity present in the original AO dataset, allowing for a more controlled evaluation of anaphoric resolution by humans and LLMs. The new datasets contain instances of Anaphoric *one* and NPE collected via Sketch Engine (Kilgariff et al. 2014) from the English Web 2021 (enTenTen21) corpus, a large-scale collection of web texts comprising over 52 billion words. We chose this corpus instead of the English Wikipedia, from which we created the original AO-NPE datasets, to include greater genre diversity. Overall, the instances included in the simplified AO datasets were shorter and structurally less complex than those in the original AO datasets, with four key adjustments:

- **Balanced Plausibility:** Both candidate antecedents are equally plausible within minimal contexts. For example, in the instance “Focusing on E* Applications leads to new requirements as well as to new technologies or extensions of existing *ones*, (<http://www2007.org/cfp-EA.php>)” both antecedent options (‘requirements’ vs. ‘technologies’) were equally probable and reasonable.
- **Consistent Grammatical Category:** As illustrated in Table F.1, in the simplified controlled AO dataset, all instances of Anaphoric *one*, specifically the count noun *one*, function as the head of noun phrases (NPs) and are accompanied by appropriate modifiers such as adjectives and determiners. We define *one*-licensors as the elements that precede *one*. Adhering to the principle that syntactic distribution determines syntactic category, the dataset includes only cases in which *one* unambiguously functions as a noun (N), serving as a proper subpart of a larger determiner phrase (DP) in which nouns remain syntactically distinct from other categories. Licensors in the dataset are restricted to determiners and adjectives. This restriction is motivated by structural considerations: if *one* were to function as the head of the DP, the presence of an additional determiner would be ruled out, since a single phrase can host only one head responsible for reference and interpretation. We therefore ensure that all instances of *one* in the dataset behave like regular count nouns, as shown by their compatibility with numerals and other determiners.
- **Strict Number Agreement:** Number agreement is consistently enforced in the simplified AO-NPE datasets. Specifically, in the original AO dataset, we intentionally included number matching and mismatching antecedent options to explore the extent to which anaphoric resolution is influenced by such morphological cues. In contrast, in the simplified AO dataset, both antecedent options strictly agree with the Anaphoric *one/s* in number. For instance, the antecedent (either correct or incorrect candidate) of a [+singular] *one* must be a [+singular] NP, and the antecedent (whether the correct or incorrect candidate) of a [+plural] *ones* must be a [+plural] NP.
- **Avoided Verb Repetition:** In the simplified AO dataset, we did not include instances in which the verb preceding the Anaphoric *one* was identical to the verb preceding the antecedent, thereby preventing models from relying on simple verb repetition patterns.

1.1 Examples from the Simplified AO corpus

Table F.1

Examples from the Simplified AO corpus.

Column 1	2	3	4	5	6	7	8	9	10	11	12
ID	text	one/ ones	antecedent type	A	B	A-antecedent	B-antecedent	A-antecedent	B-antecedent	one-licensor category	URL
1	Focusing on E* Applications leads to new requirements as well as to new technologies or extensions of existing ones .	ones	linguistic	requirements	technologies	FALSE	TRUE	non-local	local	adj	http://www.w2007.org/cfp-EA.php
2	As new components are introduced into the system, an upgrade of existing services, even the introduction of new ones not previously considered, may be needed as well.	ones	linguistic	services	components	TRUE	FALSE	local	non-local	adj	http://www.w10.org/cdrom/papers/106/
3	Using two benchmarks representative of dynamic web sites, measurements of our prototype show that it can increase end-to-end throughput by as much as twofold for non-data intensive applications and by as much as tenfold for data intensive ones .	ones	linguistic	applications	measurements	TRUE	FALSE	local	non-local	adj	http://www.w2007.org/paper120.php
4	This shift follows the noted global decline of social movements based around redistribution of material resources and the rise of new ones demanding standing premised on cultural difference.	ones	linguistic	material resources	social movements	FALSE	TRUE	local	non-local	adj	http://hnn.us/article/155722

Note. The annotation format of the simplified AO corpus basically follows that of the original GAO corpus, in which a total of 12 columns are included:

- **Column 1:** Marks the text number. "1" means that the text is the first example in the corpus. The above example lists the first 4 instances in the corpus.
- **Column 2:** Presents the raw text collected from the enTenTen21 corpus.
- **Column 3:** Marks the Anaphoric *one*. Depending on the number feature, it can be either *one* or *ones*.
- **Column 4:** Marks the type of the antecedent (either linguistic or discourse; (Hankamer and Sag 1976; Sag and Hankamer 1984)) of the Anaphoric *one*. "**Linguistic**" means that the antecedent appears within the same sentence as the Anaphoric *one*, and "**discourse**" means that the antecedent is found in the preceding context. In the simplified AO dataset, among the 150 items, 100 items (66.7%) contain a linguistic antecedent and 50 items (33.3%) contain a discourse antecedent.
- **Column 5:** Marks the first antecedent option. This antecedent candidate can be the correct or incorrect antecedent of the Anaphoric *one*, which is revealed in column 7.
- **Column 6:** Marks the second antecedent option. The TRUE/FALSE value of this antecedent candidate is revealed in column 8.
- **Column 7:** Marks whether antecedent option A is the correct or incorrect antecedent of the Anaphoric *one*.
- **Column 8:** Marks whether antecedent option B is the correct or incorrect antecedent of the Anaphoric *one*.
- **Column 9:** Marks the linear distance between antecedent option A and the Anaphoric *one*. "**Local**" means that **antecedent A** is **closer** to the Anaphoric *one* in linear distance than antecedent B, as seen in texts **ID 2, 3, 4**. In contrast, "**non-local**" means that **antecedent A** is **farther** from the Anaphoric *one* in linear distance than antecedent B, as seen in texts **ID 1**.
- **Column 10:** Marks the linear distance between antecedent option B and the Anaphoric *one*. "**Local**" means that **antecedent B** is **closer** to the Anaphoric *one* in linear distance than antecedent A, as seen in texts **ID 1**. In contrast, "**non-local**" means that **antecedent B** is **farther** from the Anaphoric *one* in linear distance than antecedent A, as seen in texts **ID 2, 3, 4**. In the simplified AO corpus, linear distance between *one* and its antecedent is counterbalanced: in half of the sentences, the correct antecedent appears closer to the Anaphoric *one* than the incorrect candidate, while in the other half, the incorrect candidate is closer.
- **Column 11:** Marks the category of the phrase that introduces the Anaphoric *one/s* (a.k.a. the licensor). In the simplified AO dataset, it is either an adjective (adj) or a determiner (det).

- **Column 12:** Presents the URL of the raw text.

1.2 Examples from the Simplified NPE corpus

Corresponding to the simplified AO dataset, we also created a simplified NPE dataset using the same enTenTen21 corpus and annotation criteria. In the simplified NPE dataset, the NPE instances are generally shorter, offer equally plausible antecedent options, and are structurally simpler than those in the GNPE corpus.

Table F.2

Examples from the Simplified NPE corpus.

column 1	2	3	4	5	6	7	8	9	10	
ID	text	licensor	A	B	A-antecedent	B-antecedent	A-antecedent	B-antecedent	NPE-licensor category	URL
3	There must be loads of anecdotes from those days, but just at the moment I'm struggling to think of any !	any	days	anecdotes	FALSE	TRUE	local	non-local	det	http://www.c64.com/gt_display_interview.php?interview=17
4	And that's where all the loose ends of the conventional novel start to come together; all the flaws of Never Let Me Go are material for a bigger metaphor than any that can be found solely in the weave of the novel.	any	metaphor	material	TRUE	FALSE	local	non-local	det	http://www.qlrs.com/critique.asp?id=441
5	The Speaker must examine all motions relating to Lords amendments at each CCLA stage, and certify any that meet the two-part test.	any	amendments	motions	FALSE	TRUE	local	non-local	det	http://evel.uak/how-does-evel-work/
6	Saiyans seem to age more gracefully than humans and any certainly look good, but it's unclear if there are any that are hundreds of years old.	any	Saiyans	humans	TRUE	FALSE	non-local	local	det	https://www.cbr.com/dragon-ball-namekians-better-than-saiyans/

Note. The annotation format of the simplified NPE corpus largely follows that of the original GNPE corpus, in which a total of 11 columns are included:

- **Column 1:** Marks the text number. "1" means that the text is the first example in the corpus. The above example lists the first four instances in the corpus.
- **Column 2:** Presents the raw text collected from the enTenTen21 corpus.
- **Column 3:** Marks the word that introduces the NPE (i.e., the licensor).
- **Column 4:** Marks the first antecedent option. This antecedent candidate can be the correct or incorrect antecedent of the NPE, which is revealed in column 6.
- **Column 5:** Marks the second antecedent option. The TRUE/FALSE value of this antecedent candidate is revealed in column 7.
- **Column 6:** Marks whether antecedent option A is the correct or incorrect antecedent of the NPE.
- **Column 7:** Marks whether antecedent option B is the correct or incorrect antecedent of the NPE.
- **Column 8:** Marks the linear distance between antecedent option A and the NPE licensor. "Local" means that **antecedent A** is **closer** to the NPE licensor in linear distance than antecedent B. In contrast, "non-local" means that **antecedent A** is **farther** from the NPE licensor in linear distance than antecedent B.
- **Column 9:** Marks the linear distance between antecedent option B and the NPE licensor. "Local" means that **antecedent B** is **closer** to the NPE licensor in linear distance than antecedent A. In contrast, "non-local" means that **antecedent B** is **farther** from the NPE licensor in linear distance than antecedent A. In the simplified NPE corpus, the linear distance between the NPE licensor and its antecedent is counterbalanced: in half of the sentences, the correct antecedent appears closer to the NPE licensor than the incorrect candidate, while in the other half, the incorrect candidate is closer.

- **Column 10:** Marks the category of the phrase that introduces the NPE (a.k.a. the licenser). In the simplified NPE dataset, it is basically a determiner (det).
- **Column 11:** Presents the URL of the raw text.

2. Results from the Simplified AO-NPE datasets, in which items were carefully selected to ensure that both antecedent candidates had comparable plausibility, that the number feature of *one/s* matched that of the antecedent, and that *one* functioned as an NP head, are shown in Table F.3.

Table F.3

Accuracy of new datasets.

	Forced-choice				Open-ended			
	LLaMA	Gemma	GPT	Human	LLaMA	Gemma	GPT	Human
One	48.67%	94%	96%	85.5%	55.33%	74%	82.67%	83.78%
NPE	49.33%	86.67%	94.67%	83.78%	58.67%	82.67%	84.67%	82.70%

The simplified AO-NPE datasets were tested with both LLMs and human participants (N = 60 native speakers of English, recruited via Prolific). All other aspects of the setup were the same as in the forced-choice and open-ended experiments. The results from the simplified AO-NPE datasets in the open-ended identification experiment revealed a pattern similar to that observed in the original AO-NPE datasets: LLMs generally performed better on NPE resolution than on Anaphoric *one* resolution, whereas humans showed the opposite trend. Furthermore, GPT outperformed the other two LLMs, while LLaMA performed the worst; overall, model performance was notably higher on these simplified datasets than on the original datasets.

The improved performance on the simplified datasets is likely attributable to their deliberate control of many linguistic factors. The resulting datasets are more balanced and simpler, as the items are shorter, less complex, and designed to maintain balanced plausibility and strict number agreement. Such simplification reduces difficulties arising from morphosyntactic mismatches and complex discourse contexts, allowing models, particularly GPT and Gemma, to employ surface cues more effectively and thus achieve higher forced-choice accuracy. By contrast, our original AO-NPE datasets reflect the natural distribution of anaphoric *one* and NPE in real-world data (Gao et al. 2023; Johnson, Kaffee, and Redi 2024), where number mismatches, DP and NP-like *one* variations, and the challenge of identifying referential antecedents in complex discourse are common. Despite the differences in performance, the qualitative patterns that human performance were more robust than models remain consistent across both the original and simplified AO-NPE datasets. This interpretation remains tentative and invites further investigation into how humans and LLMs resolve complex anaphora and how different task framings influence this process. The simplified AO-NPE datasets with fine-grained annotations and experimental results on these datasets are available in the supplementary material at <https://github.com/nayoun1/Anaphoric-One-and-NPE>.

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